

복사전달 특강

Special Topics in Radiative Transfer

Lecture 4

2026 March 24 (Tuesday), 1PM

updated on 03/24

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KASI / UST

Atomic Structure - References

- References for the atomic/molecular spectroscopy

Astronomical Spectroscopy [Jonathan Tennyson]

Physics and Chemistry of the Interstellar Medium [Sun Kwok]

Astrophysics of the Diffuse Universe [Michael A. Dopita & Ralph S. Sutherland]

⇒ many typos

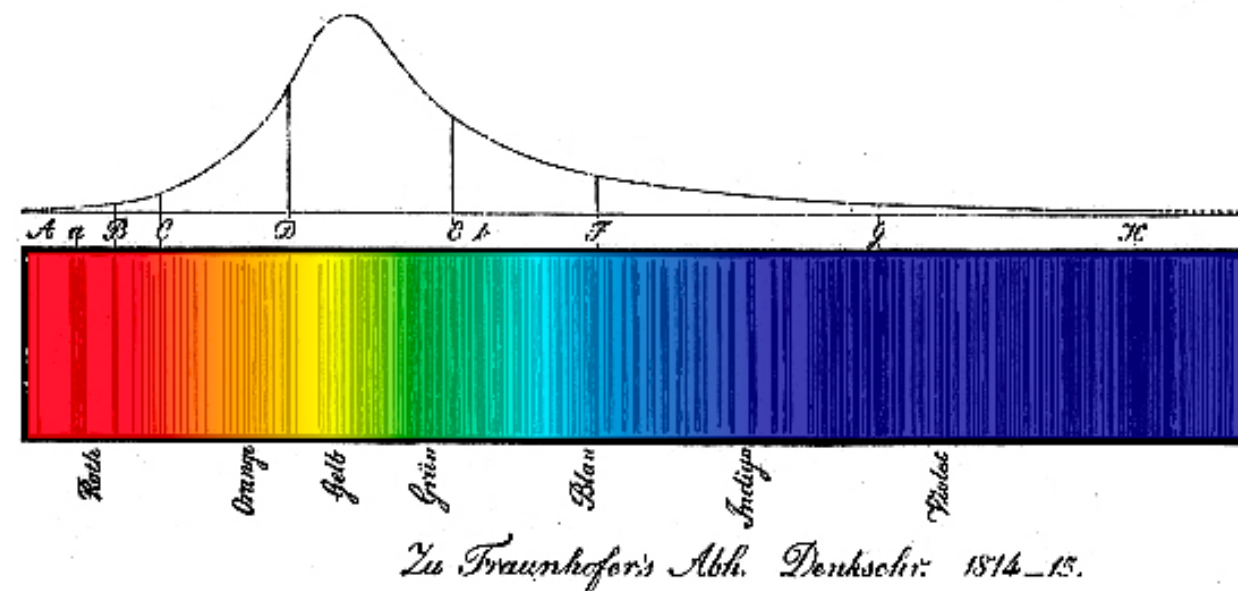
Atomic Spectroscopy and Radiative Processes [Egidio Landi Degl'Innocenti]

Atomic Astrophysics and Spectroscopy [Pradhan, A. K., & Nahar, S. N.]

History: Fraunhofer Lines

- In 1814, Joseph von Fraunhofer (1789-1826) used one of the high-quality prisms he had manufactured to diffract a beam of sunlight onto a whitewashed wall.
 - Besides the characteristic colors of the rainbow, he saw **many dark lines**.
 - He catalogued the exact wavelength of each dark line and labelled the strongest of them with letters. These are still known today as Fraunhofer lines. Many of these labels, such as the sodium D lines ($5896\text{\AA}$, $5890\text{\AA}$; Na I D₁, D₂) are still used today.
 - He did not know what caused the dark lines he observed.
 - However, he performed a similar experiment using light from the nearby star Betelgeuse and found that the pattern of dark lines changed significantly. He concluded correctly that **most of those features were somehow related to the composition of the object**.
- The first real step in understanding Fraunhofer's observations came in the middle of the 19th century with the experiments of Gustav Kirchhoff (1824-1887) and Robert Bunsen (1811-1891). They studied the color of the light emitted when metals were burnt in flames. In certain cases, the wavelength of the emitted light gave an exact match with the Fraunhofer lines. **These experiments demonstrated that the Fraunhofer lines were a direct consequence of the atomic composition of the Sun.**

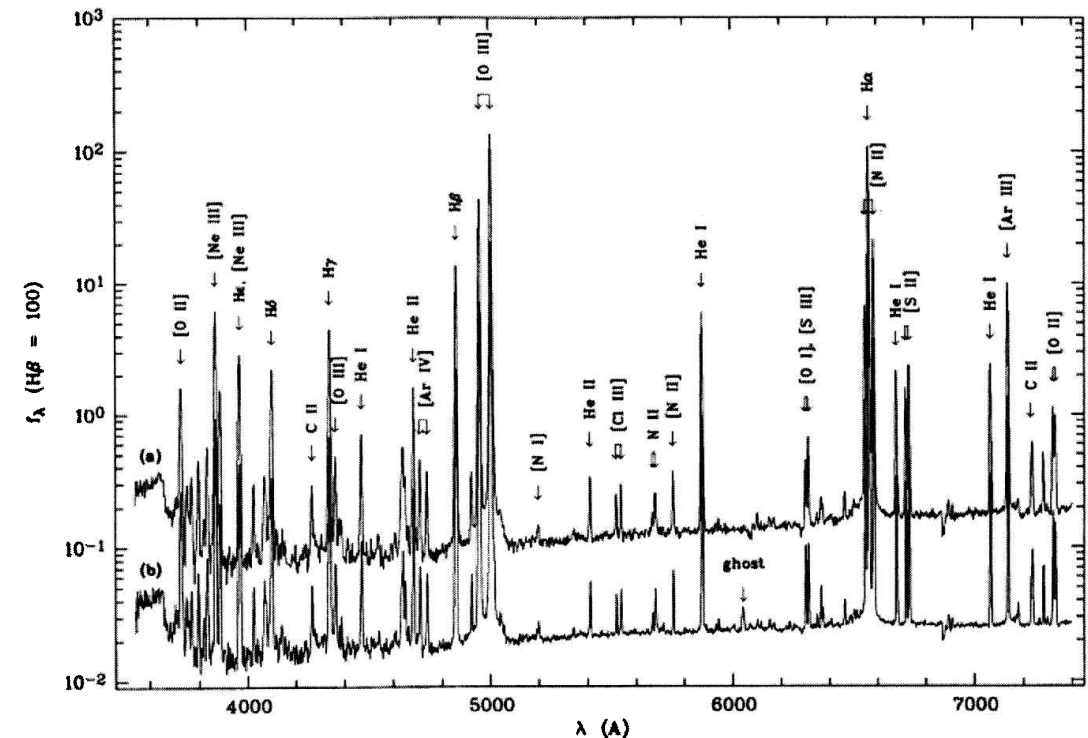
The solar spectrum as recorded by Fraunhofer (color overlaid).



In fact, some of the lines were due to the Earth's atmosphere, the so-called telluric lines.

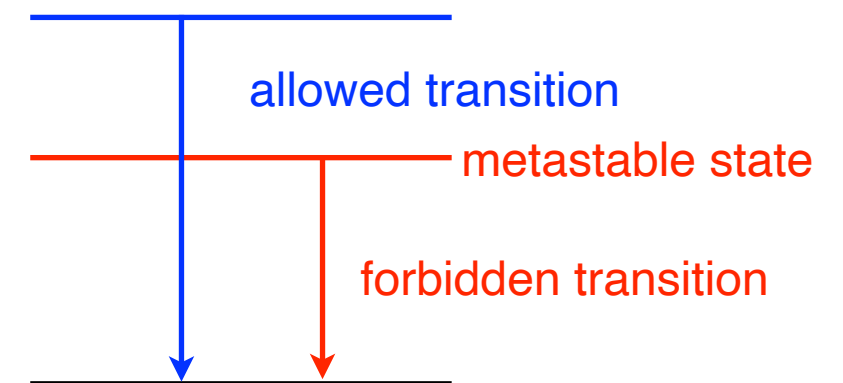
History: Nebulium?

- In 1918, extensive studies of the emission spectra of nebulae found a series of lines which had not been observed in the laboratory.
 - Particularly strong were features at 4959Å and 5007Å. For a long time, this pair could not be identified and these lines were attributed to a new element, '*nebulium*'.
 - In 1927, Ira Bowen (1898-1973) discovered that the lines were not really due to a new chemical element but instead *forbidden lines from doubly ionized oxygen* [O III].
 - He realized that in the diffuse conditions found in nebulae, atoms and ions could survive a long time without undergoing collisions. Indeed, *under typical nebula conditions the mean time between collisions is in the range 10-10,000 secs.* This means that *there is sufficient time for excited, metastable states to decay via weak, forbidden line emissions.*
 - The forbidden lines could not be observed in the laboratory where it was not possible to produce collision-free conditions over this long timeframe.
 - **Other 'nebulium' lines turned out to be forbidden lines** originating from singly ionized oxygen [O II] and nitrogen [N II].



Optical spectra of NGC 6153, Liu et al. (2000, MNRAS)

[O III], [O II], [N II], etc:
We use a pair of square brackets for a forbidden line.



Quantum Mechanics

- **The classical theory of radiation is unable to treat physical processes in which the interaction between matter and radiation takes place by means of single (or a few) photons.**

We have already dealt with some elementary aspects of this interaction when we discussed the Planck law and the Einstein coefficients.

However, to really solve problems we need to find explicit expressions for the A and B coefficients or equivalents.

This must involve detailed investigation of the structure of the matter that interacts with the radiation, its energy levels, and other physical properties.

- Astrophysical spectral lines offer two important insights into the workings of our Universe.

First, they are probes of the fundamental (QM) nature of matter because they originate from subatomic, atomic and molecular systems.

Second, they provide, via the Doppler effect, critical dynamical information on astrophysical systems ranging in scale from planetary system to superclusters of galaxies.

Schrödinger Equation

- A good starting point for a quantum mechanical understanding of spectral lines is the hydrogen atom, which is the simplest of all atoms, consisting of one proton and one electron.
- Since the H atom consists of only a proton and an electron, the Coulomb field experienced by the electron is spherically symmetric. The time-dependent Schrödinger equation for a system with **Hamiltonian H**:

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

Often we are interested in the stationary solutions found by separating the time and space parts of the wave function, which is possible *if H is independent of time*:

- The time-independent Schrödinger equation is obtained as follows:

$\Psi(\mathbf{r}, t) = \psi(\mathbf{r})e^{iEt/\hbar} \longrightarrow H\psi(\mathbf{r}) = E\psi(\mathbf{r})$ $H = \frac{p^2}{2m} + V(r) = -\frac{\hbar^2}{2m}\nabla^2 + V(r)$	momentum operator: $\mathbf{p} = \frac{\hbar}{i}\nabla$
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Here, in the spherical coordinates, the Laplace operator is given by

$$\begin{aligned} \nabla^2 &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\ &= \mathcal{R} + \frac{1}{r^2} \mathcal{L}^2 \end{aligned}$$



Erwin Schrödinger

- Separation of variables: Since $V(r)$ depends only on r , we can try something like

$$\phi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

Then, the Schrödinger equation becomes

$$\left[-\frac{\hbar^2}{2m} \mathcal{R}R(r) \right] Y(\theta, \phi) - \frac{\hbar^2}{2mr^2} [\mathcal{L}^2 Y(\theta, \phi)] R(r) + V(r)R(r)Y(\theta, \phi) = ER(r)Y(\theta, \phi)$$

$$\longrightarrow \frac{r^2 \mathcal{R}R(r)}{R(r)} + \frac{2mr^2}{\hbar^2} [E - V(r)] = -\frac{\mathcal{L}^2 Y(\theta, \phi)}{Y(\theta, \phi)} = \Lambda \quad (\text{a constant})$$

$$\longrightarrow \begin{aligned} \mathcal{R}R(r) + \frac{2m}{\hbar^2} [E - V(r)] R(r) &= \frac{\Lambda}{r^2} R(r) \\ \mathcal{L}^2 Y(\theta, \phi) &= -\Lambda Y(\theta, \phi) \end{aligned}$$

The left side depends only on r , while the right side on θ and ϕ . Therefore, they should be a constant.

The solution of the second equation is known to be the spherical harmonics.

$$Y(\theta, \phi) = Y_{\ell m}(\theta, \phi) \quad \text{and} \quad \Lambda = \ell(\ell + 1)$$

The solution for the radial component is given by the associated Laguerre polynomials.

$$R_{n\ell}(r) = -2 \left(\frac{1}{na_0} \right)^{3/2} \sqrt{\frac{(n - \ell - 1)!}{n [(n + \ell)!]^3}} \left(\frac{2r}{na_0} \right)^\ell e^{-r/na_0} L_{n+\ell}^{2\ell+1} \left(\frac{2r}{na_0} \right)$$

Summary of the Hydrogen atom

- momentum operator $\mathbf{p} = \frac{\hbar}{i} \nabla$
- Hamiltonian operator $H = \frac{p^2}{2m} + V = -\frac{\hbar^2}{2m} \nabla^2 + V$
- Expectation value of an operator

$$\langle F \rangle = \int \psi^* F \psi d^3x \quad \rightarrow \quad \langle F \rangle = \langle \psi | F | \psi \rangle$$

- Wavefunction for a hydrogen-like atom:

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_{lm}(\theta, \phi)$$

radial function:

$$R_{nl}(r) = - \left[\left(\frac{2Z}{na_0} \right)^3 \frac{(n-l-1)!}{2n\{(n+l)!\}^3} \right]^{1/2} e^{-\rho/2} \rho^l L_{n+l}^{2l+1}(\rho)$$

spherical harmonics

$$Y_{lm}(\theta, \phi) = (-1)^{(m+|m|)/2} \left[\frac{(l-|m|)!}{(l+|m|)!} \frac{2l+1}{4\pi} \right]^{1/2} P_l^{|m|}(\cos \theta) e^{im\phi}$$

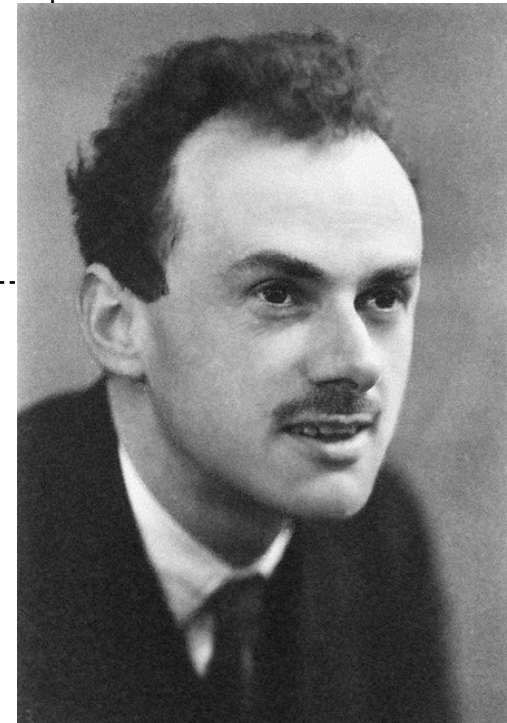
- $n = 1, 2, 3, \dots$: **principal quantum number**
- $l = 0, 1, 2, \dots, n-1$: **orbital angular momentum quantum number**
- $m = -l, -l+1, \dots, 0, \dots, l-1, l$: **magnetic quantum number**

bra-ket notation (Dirac notation)

ket vector : $|\psi\rangle$

bra vector : $\langle\phi| = |\psi\rangle^\dagger$

($\dagger$: conjugate transpose)



Paul Adrien Maurice Dirac

$$\rho = \frac{2Z}{na_0} r, \quad a_0 \equiv \frac{\hbar^2}{m_e c^2} = 0.529 \text{Å (Bohr radius)},$$

L_{n+l}^{2l+1} = associated Laguerre polynomial

$$P_l^{|m|} = \text{associated Legendre function}$$

Quantum Numbers / H-atom

- Each bound state of the hydrogen atom is characterized by a set of four quantum numbers (n, l, m, m_s)

- $n = 1, 2, 3, \dots$: principal quantum number (shell)
- $l = 0, 1, 2, \dots, n - 1$: **orbital angular momentum** quantum number (subshell)
 - By convention, the values of l are usually designated by small letters.

0	1	2	3	4	5	6	7	8	...
s	p	d	f	g	h	i	k	l	...

- $m = -l, -l + 1, \dots, 0, \dots, l - 1, l$: magnetic quantum number.
 - It determines the behavior of the energy levels in the presence of a magnetic field.
 - This is the projection of the electron orbital angular momentum along the z -axis of the system.
- Spin
 - The electron possesses an intrinsic, **spin angular momentum** with the magnitude of $|s| = \frac{1}{2}$.
 - There are two states, $m_s = \pm \frac{1}{2}$, for the spin.
- Degeneracy for a given n : $2 \times \sum_{l=0}^{n-1} (2l + 1) = 2n^2$

- Wavefunctions for a hydrogen-like atom

$$R_{1,0} = \left(\frac{Z}{a_0} \right)^{3/2} 2 e^{-\rho}$$

$$R_{2,0} = \left(\frac{Z}{2a_0} \right)^{3/2} 2 (1 - \rho) e^{-\rho}$$

$$R_{2,1} = \left(\frac{Z}{2a_0} \right)^{3/2} \frac{2}{\sqrt{3}} \rho e^{-\rho}$$

$$R_{3,0} = \left(\frac{Z}{3a_0} \right)^{3/2} 2 \left(1 - 2\rho + \frac{2}{3}\rho^2 \right) e^{-\rho}$$

$$R_{3,1} = \left(\frac{Z}{3a_0} \right)^{3/2} \frac{4\sqrt{2}}{3} \rho \left(1 - \frac{1}{2}\rho \right) e^{-\rho}$$

$$R_{3,2} = \left(\frac{Z}{3a_0} \right)^{3/2} \frac{2\sqrt{2}}{3\sqrt{5}} \rho^2 e^{-\rho}$$

Normalisation: $\int_0^\infty R_{n,l}^2 r^2 dr = 1$

$$Y_{0,0} = \sqrt{\frac{1}{4\pi}}$$

$$Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

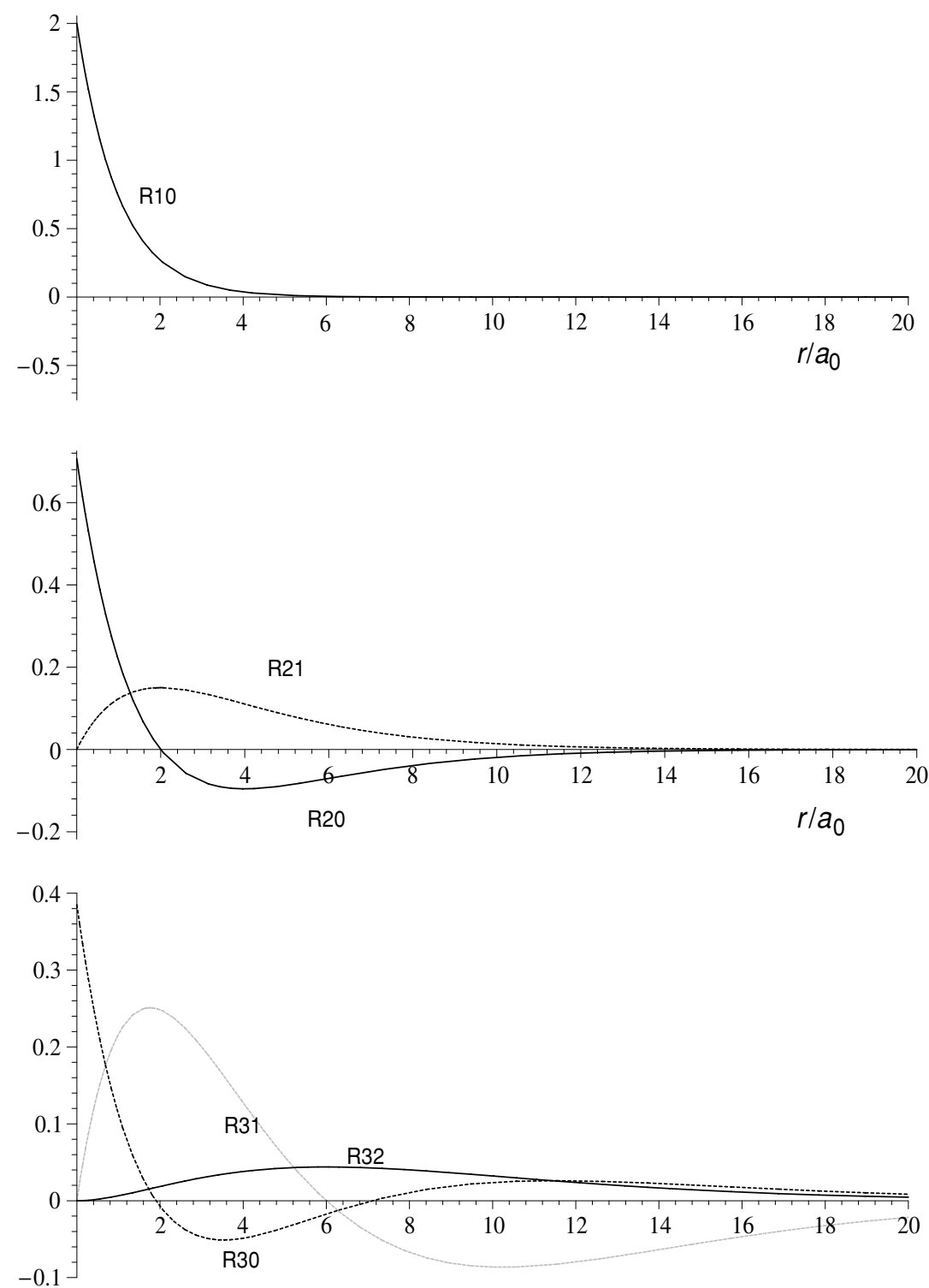
$$Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}$$

$$Y_{2,0} = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$$

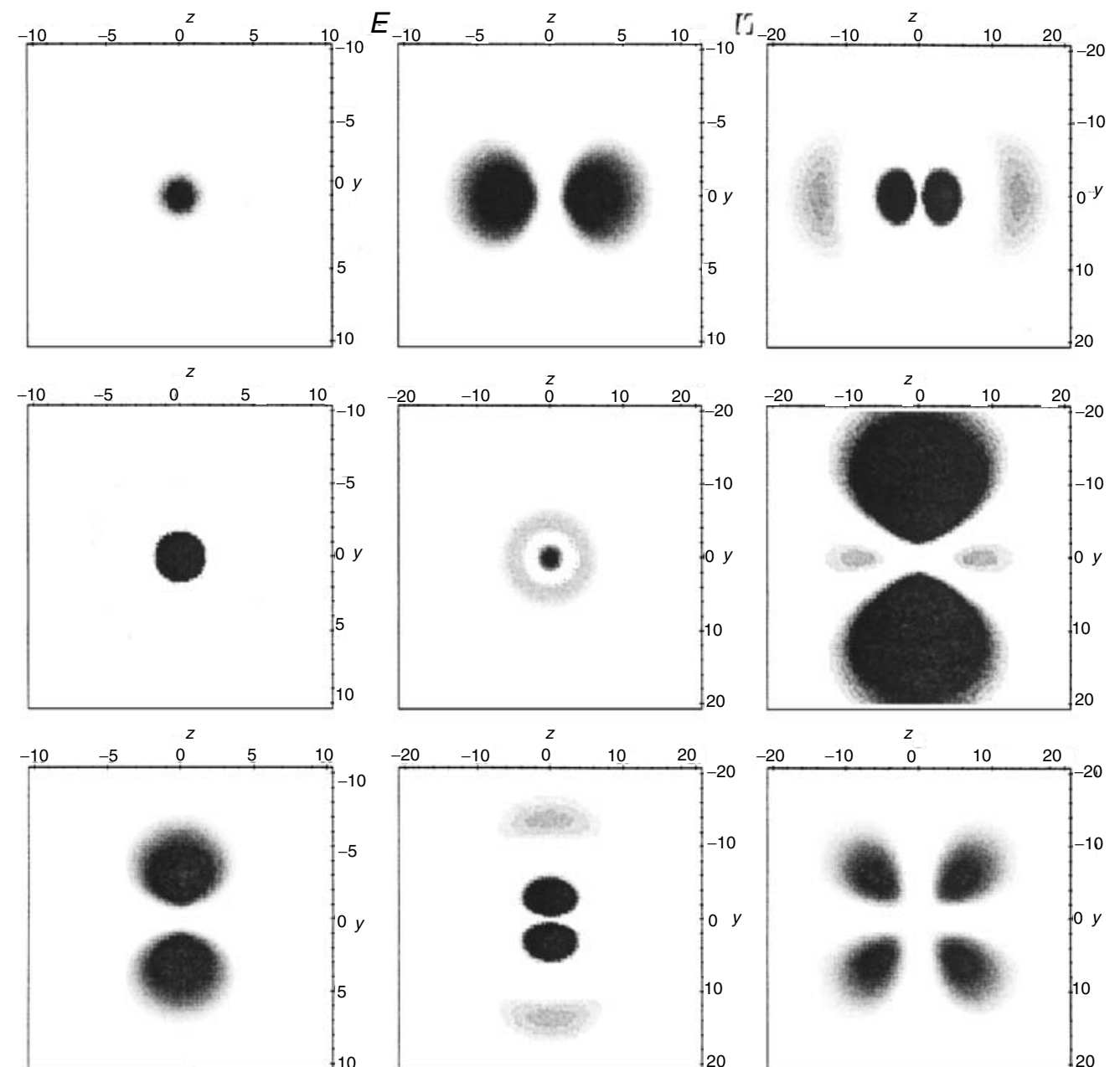
$$Y_{2,\pm 1} = \mp \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\phi}$$

$$Y_{2,\pm 2} = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\phi}$$

Normalisation: $\int_0^{2\pi} \int_0^\pi |Y_{l,m}|^2 \sin \theta d\theta d\phi = 1$



Radial wave functions for the first principal quantum numbers of hydrogen



Probability distributions (per unit volume). A slice through the $\phi = 0$ plane. The darker the shading the greater the probability density.

[Left] 1s, 2s, 2p0 from top to bottom

[Center] 2p1, 3s, 3p0

[Right] 3p1, 3d0, 3d1

[Angular Momentum]

- The orbital angular momentum operator $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ has the following eigenvalue.

$$\mathbf{L}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}, \quad L_z Y_{lm} = m\hbar Y_{lm}$$

Here, the ℓ can have values $\ell = 0, 1, \dots, n-1$.

The magnitude of the angular momenta are

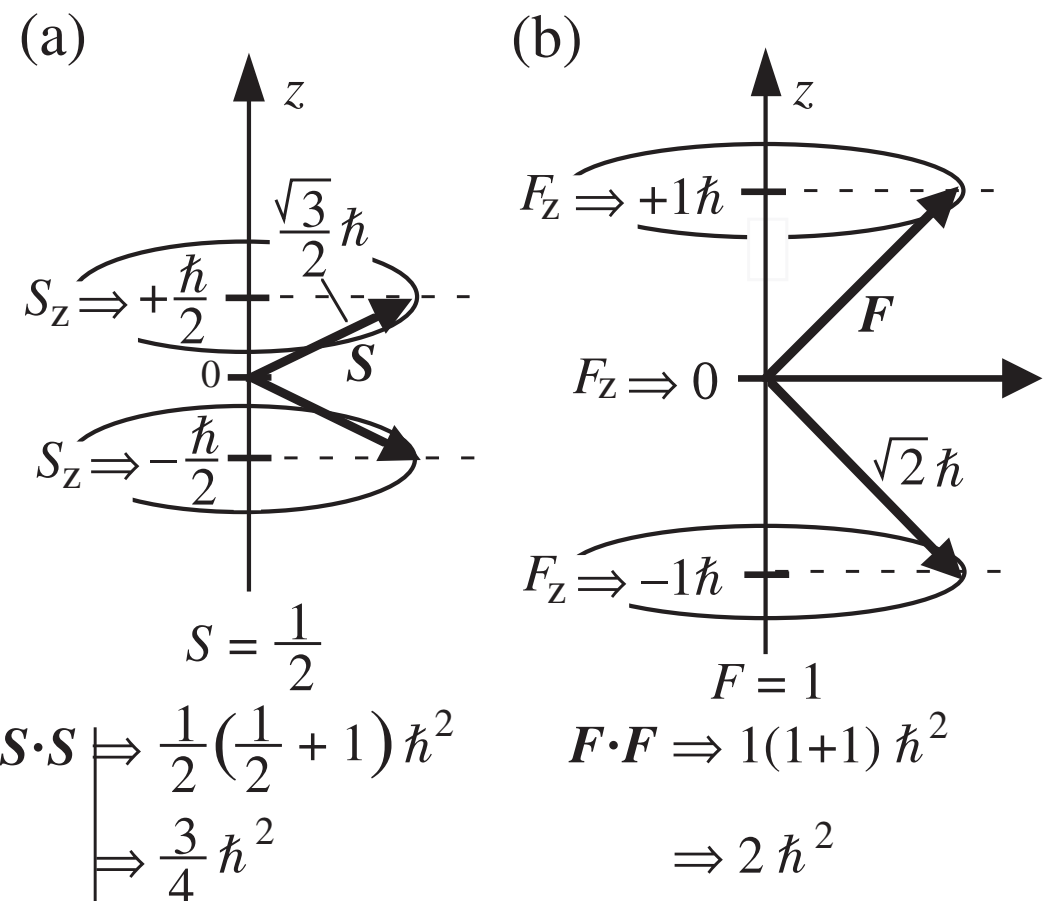
$$|\mathbf{L}| = \sqrt{\ell(\ell+1)}\hbar \quad \text{where } \ell = 0, 1, \dots, n-1$$

The z -component have the following values.

$$L_z = m\hbar \quad \text{where } m = -\ell, -\ell+1, \dots, \ell-1, \ell$$

- Electron has a spin of $s = \frac{1}{2}$ and it has two z -components:

$$s = \frac{1}{2}\hbar \rightarrow m_s = -\frac{1}{2}\hbar, \frac{1}{2}\hbar$$



- The spherical harmonics are eigenfunctions of the orbital angular momentum operator $\mathbf{L} = \mathbf{r} \times \mathbf{p}$.

$$\mathbf{L}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}, \quad L_z Y_{lm} = m\hbar Y_{lm}$$

The sizes of the angular momentum and z -component are

$$|\mathbf{L}| = \sqrt{l(l+1)}\hbar, \quad |L_z| = m\hbar$$

- Wavefunction (with $l = 0$) at $r = 0$

$$R_{10}(r) = \frac{2}{(a_0/Z)^{3/2}} e^{-r/(a_0/Z)}$$

$$R_{20}(r) = \frac{1}{(2a_0/Z)^{3/2}} \left(2 - \frac{r}{a_0/Z} \right) e^{-r/(2a_0/Z)} \longrightarrow$$

$$Y_{lm}(\theta, \phi) = \sqrt{\frac{1}{4\pi}}$$

$$|\psi_{n,l=0}(0)|^2 = \frac{1}{\pi} \left(\frac{Z}{na_0} \right)^3 \neq 0$$

This indicates that the electron can interact with the nucleus $\Rightarrow$ hyperfine structure!

Probability of being located within a volume $d^3x = |\psi(r, \theta, \phi)|^2 d^3x = R_{nl}^2(r) |Y_{lm}(\theta, \phi)|^2 r^2 \sin \theta dr d\theta d\phi$

- Properties of the spherical harmonics:

$$\int_0^\pi \int_0^{2\pi} |Y_{lm}(\theta, \phi)|^2 \sin \theta d\theta d\phi = 1$$

$$\sum_{m=-l}^l |Y_{lm}(\theta, \phi)|^2 = \frac{2l+1}{4\pi}$$

No dependency on ϕ .

This property implies that **closed shells are spherically symmetric and have very little interaction with external electrons.**

- Orthonormality
$$\int d\Omega Y_{\ell m}^*(\theta, \phi) Y_{\ell' m'}(\theta, \phi) = \delta_{\ell, \ell'} \delta_{m, m'}$$
$$\int_0^\infty R_{n\ell}(r) R_{n'\ell'}(r) dr = \delta_{n, n'}$$

- Energy

$$E_n = -Z^2(13.6 \text{ eV})/n^2$$

- Degeneracy for a given n .

$$2 \times \sum_{l=0}^{n-1} (2l+1) = 2n^2$$

$n = 1, 2, 3, \dots$: principal quantum number

$l = 0, 1, 2, \dots, n-1$: orbital angular momentum quantum number

$m = -l, -l+1, \dots, 0, \dots, l-1, l$: magnetic quantum number

- Spin

The electron possesses an intrinsic angular momentum with the magnitude of $|s| = \frac{1}{2}$.

There are two states, $m_s = \pm \frac{1}{2}$, for the spin.

Wavefunctions: $\left| \frac{1}{2} \right\rangle \equiv \alpha = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \left| -\frac{1}{2} \right\rangle \equiv \beta = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

To incorporate spin into the theory in a completely satisfactory way, one should use the relativistic Dirac equation.

H-atom Spectra

- Spectral series of the H atom
 - The spectrum of H is divided into a number of series linking different upper levels n_2 with a single lower level n_1 value. ***Each series is denoted according to its n_1 value and is named after its discoverer.***
 - Within a given series, ***individual transitions are labelled by Greek letters.***

$n_2 \longleftrightarrow n_1$			
n_1	Name	Symbol	Spectral region
1	Lyman	Ly	ultraviolet
2	Balmer	H	visible
3	Paschen	P	infrared
4	Brackett	Br	infrared
5	Pfund	Pf	infrared
6	Humphreys	Hu	infrared

$$\Delta n \equiv n_2 - n_1$$

$$\Delta n = 1 \text{ is } \alpha,$$

$$\Delta n = 2 \text{ is } \beta,$$

$$\Delta n = 3 \text{ is } \gamma,$$

$$\Delta n = 4 \text{ is } \delta,$$

$$\Delta n = 5 \text{ is } \epsilon.$$

Lyman series : $\text{Ly}\alpha, \text{Ly}\beta, \text{Ly}\gamma, \dots$

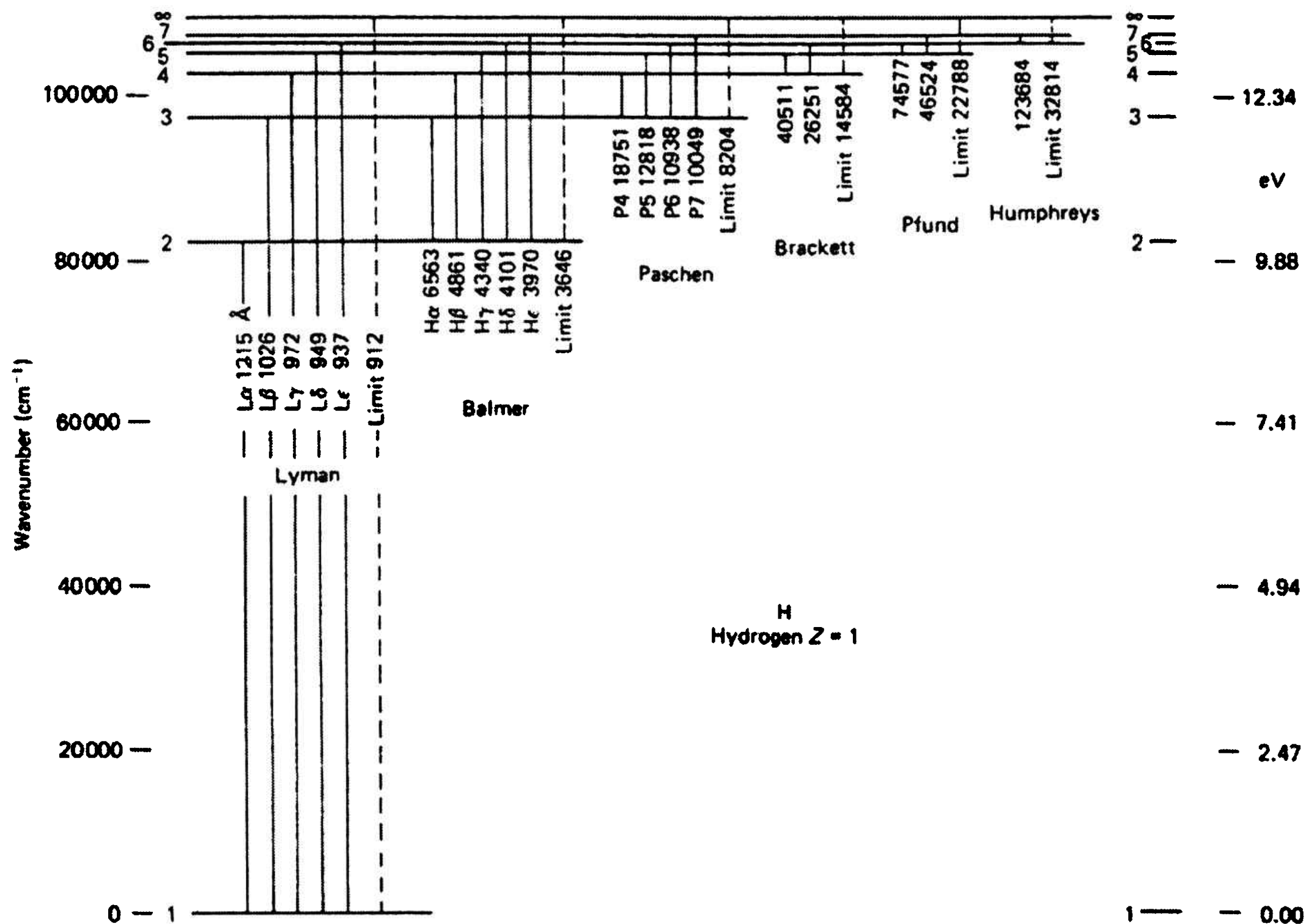
Balmer series : $\text{H}\alpha, \text{H}\beta, \text{H}\gamma, \dots$

Paschen series: $\text{P}\alpha, \text{P}\beta, \text{P}\gamma, \dots$

Brackett series : $\text{Br}\alpha, \text{Br}\beta, \text{Br}\gamma, \dots$

Transitions with high Δn are labelled by the n_2 .
Thus, $H15$ is the Balmer series transition
between $n_1 = 2$ and $n_2 = 15$.

Schematic energy levels of the hydrogen atom with various spectral series identified. The vertical numbers are wavelengths in Å.



Many electron systems

- The time-independent Schrödinger equation for an atom with N electrons and nuclear charge (atomic number) Z .

$$\left[\sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 - \frac{Ze^2}{r} \right) + \sum_{i=1}^{N-1} \sum_{j=1}^N \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \right] \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$$

where $\mathbf{r}_i$ is the coordinate of the i th electron, with its origin at the nucleus.

The first term contains a kinetic energy operator for the motion of each electron and the Coulomb attraction between that electron and the nucleus.

The second term contains the electron-electron Coulomb repulsion term.

The Coulomb repulsion between pairs of electrons means the above equation is not analytically solvable, even for the simplest case, the helium atom for which $N = 2$.

Many-Electron Systems - [Central Field Approximation]

- Even in complex atoms with N electrons it is useful to consider single-electron states.

Hartree-Fock approximation (or self-consistent field approximation): We assume that each electron moves in the potential of the nucleus plus the averaged potential due to the other $N - 1$ electrons.

Central field approximation (or orbital approximation): In addition, when this averaged potential is assumed to be spherically symmetric, the force acting on each electron only depends on its distance from the nucleus at the center. It provides a useful classification of atomic states and also a starting point.

Let us assume that each electron moves in its own (angle-independent) central potential given by $V_i(r_i)$. This gives a simplified Schrödinger equation for the motion of each electron:

$$\left[-\frac{\hbar^2}{2m} \nabla_i^2 + V_i(r_i) \right] \phi_i(\mathbf{r}_i) = E_i \phi_i(\mathbf{r}_i) \quad \text{where} \quad V_i(r_i) = -\frac{Ze^2}{r_i} + \sum_{j \neq i} \left\langle \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \right\rangle$$

Note the average sign to make the potential angle-independent.

Orbitals: The solutions of the above equation are known as orbitals.

Using this approximation, the total energy and the total wave function of the system are given by the sum of single electron energies and the product of single electron wave functions, respectively.

$$E = \sum_i E_i \qquad \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \phi_1(\mathbf{r}_1) \phi_2(\mathbf{r}_2) \cdots \phi_N(\mathbf{r}_N)$$

However, this wave function fails to satisfy a basic principle of quantum mechanics that one cannot distinguish between electron i and electron j .

[Indistinguishable Particles]

- Consider a system with **two identical particles**. We note that what is physically observable is not a wave function but a probability distribution. This distribution cannot be altered by interchanging the particles. This means that

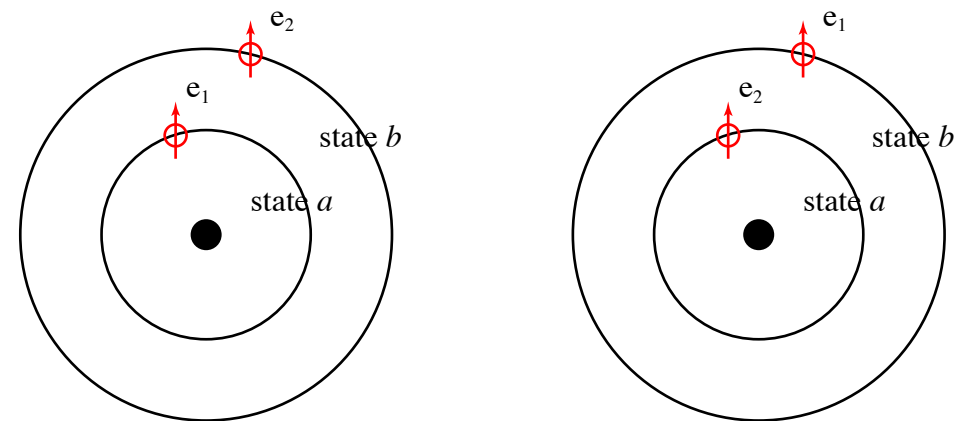
$$|\Psi(a_1, b_2)|^2 = |\Psi(a_2, b_1)|^2$$

The equation has two possible solutions:

- symmetric solution
- antisymmetric solution

$$\Psi(a_1, b_2) = \Psi(a_2, b_1)$$

$$\Psi(a_1, b_2) = -\Psi(a_2, b_1)$$



Particles 1 and 2 are indistinguishable.

Pauli Exclusion Principle: *Wave functions are antisymmetric with respect to interchange of identical Fermions* (with a half-odd-integer spin). Within the central field approximation, a two-electron wave function which obeys the Pauli Principle can be written

$$\Psi(1, 2) = \frac{1}{\sqrt{2}} [\phi_a(1)\phi_b(2) - \phi_a(2)\phi_b(1)] = -\Psi(2, 1)$$

If the two spin-orbitals are the same ($\phi_a = \phi_b$), then the total wave function is zero, i.e., $\Psi(1, 2) = 0$, and no physical (normalizable) state exists. The Pauli exclusion principle is summarized as ***“No two electrons can occupy the same spin-orbital state.”***

This exclusion provides the **degeneracy pressure** which holds up the gravitational collapse of white dwarfs and neutron stars.



Wolfgang Ernst Pauli

Complex Atoms : Electron Configuration

- The **configuration** is the distribution of electrons of an atom in atomic **orbitals**.
 - The configuration of an atomic system is defined by specifying the nl values of all the electron orbitals: nl^x means x electrons in the orbital defined by n and l .
 - Each orbital labelled nl actually consists of orbitals with $2l + 1$ different m values, each with two possible values of m_s . Thus the nl orbital can hold a maximum $2(2l + 1)$ electrons.

$$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} \dots$$

nl^x means x electrons in the orbital defined by n and l .

- shells, subshells:
 - **Principal quantum number = shell:** Shells correspond with the principal quantum numbers (1, 2, 3, ...). They are labeled alphabetically with letters used in the X-ray notation (K, L, M, ...).
 - **Orbital angular momentum quantum number = subshell:** Each shell is composed of one or more subshells. The first (K) shell has one subshell, called “1s”; The second (L) shell has two subshells, called “2s” and “2p”.

- open shell configuration, closed shell configuration:

the ground state configuration of carbon, which has six electrons: $1s^2 2s^2 2p^2$

the ground state configuration of neon atom, which has ten electrons: $1s^2 2s^2 2p^6$

A closed shell or sub-shell makes no contribution to the total orbital or spin angular momentum (L or S).

- Atomic ions which have the same number of electrons form what are called **isoelectronic series**.
- Electronically-excited states of atoms usually arise when one of the outermost electrons jumps to a higher orbital.

States with two electrons simultaneously excited are possible but are less important. For many systems, all of these states are unstable. They have sufficient energy to **autoionize** by spontaneously ejecting an electron.

[Electron Configuration]

- Energy ordering:**

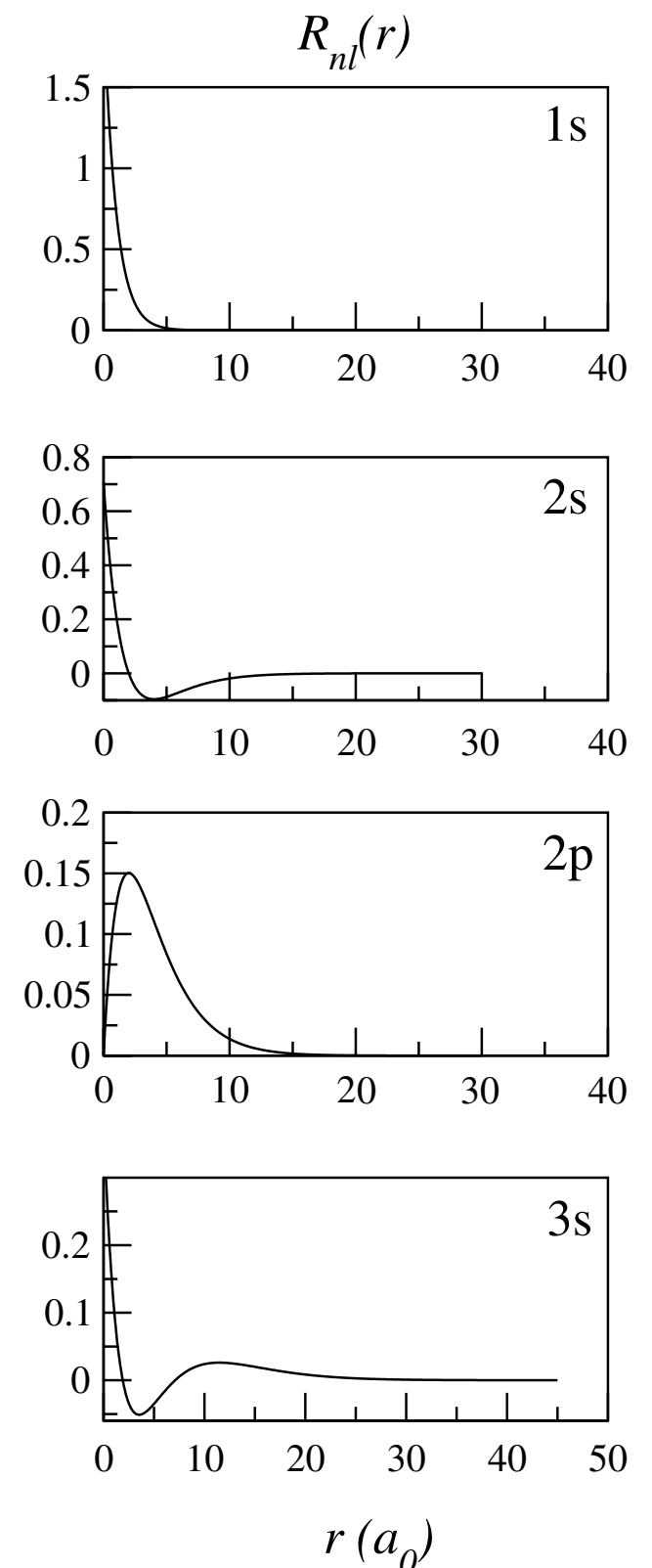
For a hydrogen-like atom, the energy of the individual orbitals is determined by principal quantum number n .

$$E(1s) < E(2s) = E(2p) < E(3s) = E(3p) = E(3d) < E(4s) \dots$$

For complex atoms, the degeneracy on the orbital angular momentum quantum number l is lifted.

This is because electrons in low l orbits ‘penetrate’, i.e., get inside orbitals with lower n -values. Penetration by the low l electrons means that they spend some of their time nearer the nucleus experiencing an enhanced Coulomb attraction. This lowers their energy relative to higher l orbitals which penetrate less or not at all.

$$E(1s) < E(2s) < E(2p) < E(3s) < E(3p) < E(4s) \lesssim E(3d) \dots$$



[Electron Configuration - Orbitals]

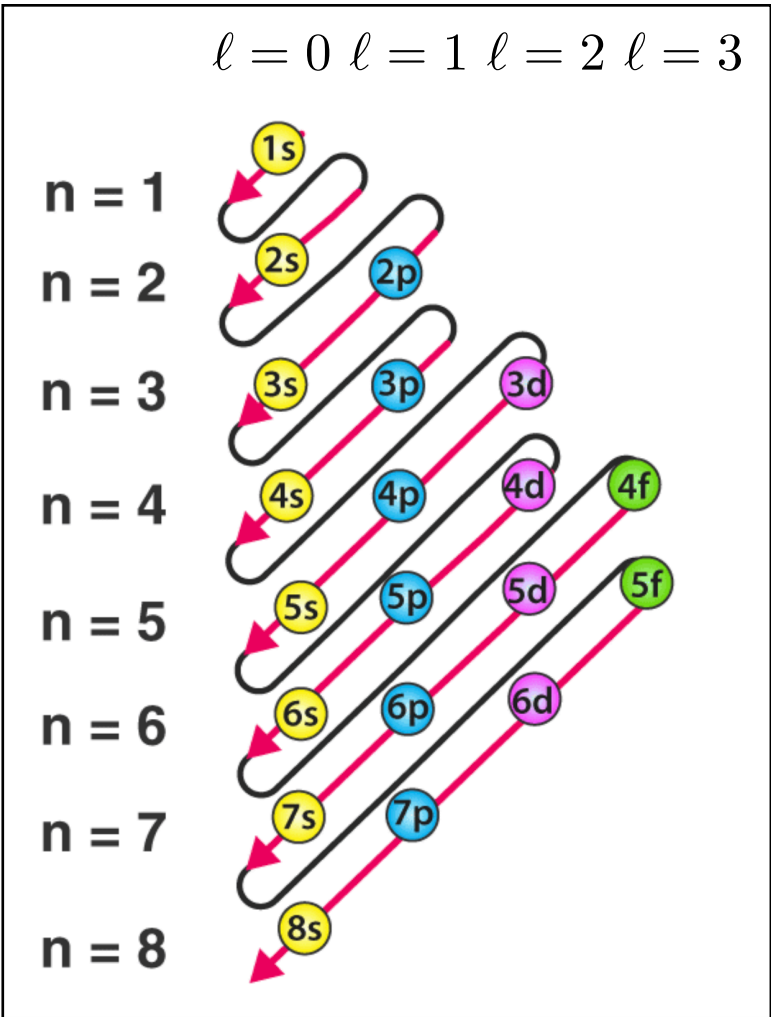
- $n = 1, 2, 3, \dots$: principal quantum number (주양자수)
- $l = 0, 1, 2, \dots, n-1$: orbital angular momentum (quantum number) for each electron (부양자수)

$n = 1, 2, 3, 4, 5 \dots$

$l = 0, 1, 2, 3, \dots, n - 1 \rightarrow s, p, d, f, g, \dots$

- The **electron configuration** represents how the electrons are filled in shells and subshells.

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 \dots$



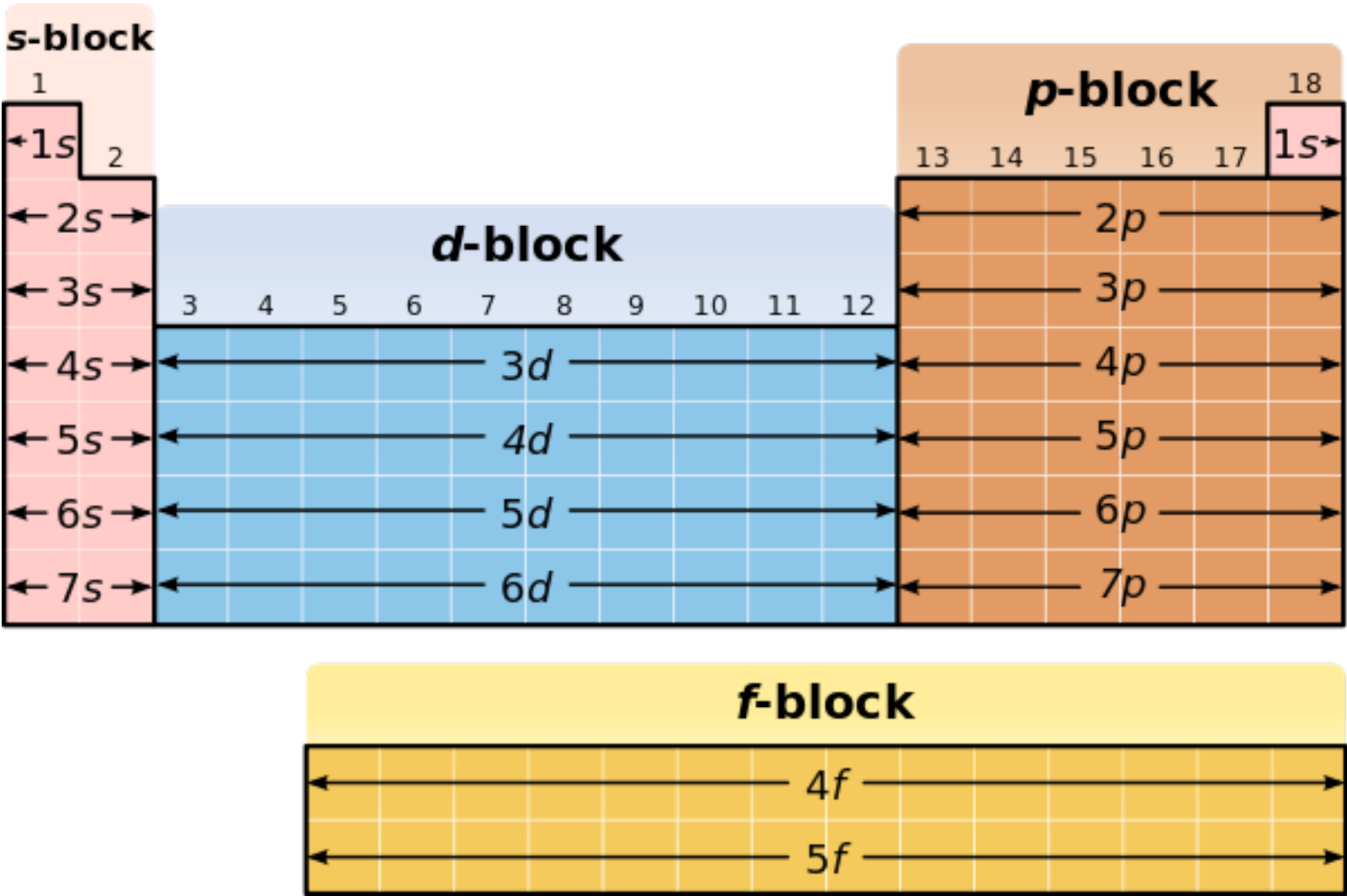
Shell (Principle Quantum Number Value)	Subshell (Angular Momentum Quantum Number)	Notation	Maximum Number of electrons
n = 1	$l=0$	1s	2
n = 2	$l=0$	2s	2
	$l=1$	2p	6
n = 3	$l=0$	3s	2
	$l=1$	3p	6
	$l=2$	3d	10
n = 4	$l=0$	4s	2
	$l=1$	4p	6
	$l=2$	4d	10
	$l=3$	4f	14

표 9.2 원자의 전자 배열과 전리 에너지(Kramida et al. 2018)

원소/ 원자번호	K		L		M			N				바닥상태 준위	전리 에너지(eV)	
	1s	2s	2p	3s	3p	3d	4s	4p	4d	4f			I → II	II → III
H	1	1									$2S_{1/2}$	13.598		
He	2	2									$1S_0$	24.587	54.418	
Li	3	2	1								$2S_{1/2}$	5.392	75.640	
Be	4	2	2								$1S_0$	9.323	18.211	
B	5	2	2	1							$2P_{1/2}^o$	8.298	25.155	
C	6	2	2	2							$3P_0$	11.260	24.385	
N	7	2	2	3							$4S_{3/2}^o$	14.534	29.601	
O	8	2	2	4							$3P_2$	13.618	35.121	
F	9	2	2	5							$2P_{3/2}^o$	17.423	34.971	
Ne	10	2	2	6							$1S_0$	21.565	40.963	
Na	11	Neon 배열			1						$2S_{1/2}$	5.139	47.286	
Mg	12				2						$1S_0$	7.646	15.035	
Al	13				2	1					$2P_{1/2}^o$	5.986	18.829	
Si	14				2	2					$3P_0$	8.152	16.346	
P	15				2	3					$4S_{3/2}^o$	10.487	19.769	
S	16				2	4					$3P_2$	10.360	23.338	
Cl	17				2	5					$2P_{3/2}^o$	12.968	23.814	
Ar	18				2	6					$1S_0$	15.760	27.630	
K	19	Argon 배열				-	1				$2S_{1/2}$	4.341	31.625	
Ca	20					-	2				$1S_0$	6.113	11.872	
Sc	21					1	2				$2D_{3/2}$	6.561	12.800	
Ti	22					2	2				$3F_2$	6.828	13.576	
V	23					3	2				$4F_{3/2}$	6.746	14.634	
Cr	24					5	1				$7S_3$	6.767	16.486	
Mn	25					5	2				$6S_{5/2}$	7.434	15.640	
Fe	26					6	2				$5D_4$	7.902	16.199	
Co	27					7	2				$4F_{9/2}$	7.881	17.084	
Ni	28					8	2				$3F_4$	7.640	18.169	
Cu	29					10	1				$2S_{1/2}$	7.726	20.292	
Zn	30					10	2				$1S_0$	9.394	17.964	
Ga	31					10	2	1			$2P_{1/2}^o$	5.999	20.515	
Ge	32					10	2	2			$3P_0$	7.899	15.935	
As	33					10	2	3			$4S_{3/2}^o$	9.789	18.589	
Se	34					10	2	4			$3P_2$	9.752	21.196	
Br	35					10	2	5			$2P_{3/2}^o$	11.814	21.591	
Kr	36					10	2	6			$1S_0$	14.000	24.360	

The subshell structure of elements up to argon ($Z = 18$) is filled up in a naturally straightforward manner, first according to n and then according to ℓ .

The 3p subshell is all occupied in argon (noble gas) with a closed subshell $3p^6$. The next element potassium (K) ($Z = 19$), begins by filling in the 4s, instead of 3d.



[The Electrostatic Interaction, LS coupling, Terms]

- The Electrostatic Interaction**

The specification of the electron configuration (the n, l vales of all electrons) leaves a great deal of unspecified information, since we are not given the values of m_l and m_s . In the central field approximation all of these states are degenerate.

The exact Hamiltonian can be written as

$$\begin{aligned}
 H &= \sum_{i=1}^N \left(\frac{P_i^2}{2m} - \frac{Ze^2}{r_i} \right) + \sum_{i=1}^N \sum_{j \neq i} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + H_{\text{so}} \\
 &= \sum_{i=1}^N \left(\frac{P_i^2}{2m} + V_i(r_i) \right) + \sum_{i=1}^N \sum_{j \neq i} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i=1}^N \sum_{j \neq i} \left\langle \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \right\rangle + H_{\text{so}} \\
 &= H_0 + H_{\text{es}} + H_{\text{so}}
 \end{aligned}$$

where $V_i(r_i) = -\frac{Ze^2}{r_i} + \sum_{j \neq i} \left\langle \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \right\rangle$

$H_0 \equiv$ central field approximation

$H_{\text{es}} \equiv \sum_{i=1}^N \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i=1}^N \sum_{j \neq i} \left\langle \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \right\rangle$

$=$ electrostatic interation

$H_{\text{so}} \equiv$ spin-orbit interation

The term H_{es} represents the residual electrostatic interaction between electrons after the averaged central field has been subtracted.

The term H_{so} is the spin-orbit interaction, which causes the fine structure splitting.

- **LS coupling (Russell-Saunders coupling): Spectroscopic Terms**

- The electrostatic interaction is the dominant splitting interaction of a configuration for many atoms (especially of low Z).
- The individual orbital angular momenta will not remain constant under this interaction, although their total $\mathbf{L} = \sum_i \mathbf{l}_i$ will be constant.

Also the sum of the spin angular momenta, $\mathbf{S} = \sum_i \mathbf{s}_i$, will be constant.

- The orbital and spin angular momenta of the electrons are added separately to give the total orbital angular momentum and the total electron spin angular momentum.
- **According to perturbation theory, the configurations split into terms with particular values of $\mathbf{L}$ and $\mathbf{S}$. These terms then split further by the action of the spin-orbit interaction.**
- Physical origin of the electrostatic splitting: The electrons repel each other, and therefore their mutual electrostatic energy is positive. *The farther away the electrons get, the lower will be the contribution of the (positive) electrostatic energy to the total energy.*
 - ♦ A large spin implies that the individual spins are aligned in the same direction. Then, the electrons will be further apart on the average by the nature of the Pauli principle. Terms with larger spin tend to lie lower in energy because of lower, positive electrostatic energies.
 - ♦ There is a similar effect regarding the orbital angular momentum. This effect is usually smaller than for the spin.

⇒ The above two properties give rise to the so called Hund's rule.

Angular Momentum Coupling

- Atoms contain several sources of angular momentum.
 - electron orbital angular momentum L
 - electron spin angular momentum S
 - nuclear spin angular momentum I
 - The nuclear spin arises from the spins of nucleons. Protons and neutrons both have an intrinsic spin of a half.
- As in classical mechanics, only the total angular momentum is a conserved quantity.
 - It is therefore necessary to combine angular momenta together.
- Addition of two angular momenta:
 - The orbital and spin angular momenta are added vectorially as $\mathbf{J} = \mathbf{L} + \mathbf{S}$. This gives the total electron angular momentum.
 - One then combines the total electron and nuclear spin angular momenta to give the final angular momentum $\mathbf{F} = \mathbf{J} + \mathbf{I}$.



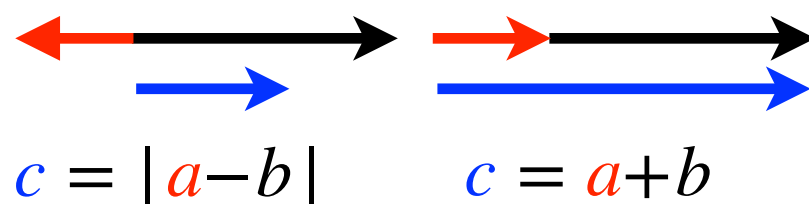
Henry Norris Russell

Hertzprung-Russell diagram
Russell-Saunders coupling

Addition of two angular momenta

- In classical mechanics, adding vector $\mathbf{a}$ and vector $\mathbf{b}$ gives a vector $\mathbf{c}$, whose length must lie in the range

$$|a - b| \leq c \leq a + b \quad \text{Here, } a, b, c \text{ are the lengths of their respective vectors.}$$



- In quantum mechanics, a similar rule applies except that the results are quantized. The allowed values of the quantized angular momentum, c , span the range from the sum to the difference of a and b in steps of one:

$$c = |a - b|, |a - b| + 1, \dots, a + b - 1, a + b$$

- For example, add the two angular momenta $L_1 = 2$ and $L_2 = 3$ together to give $\mathbf{L} = \mathbf{L}_1 + \mathbf{L}_2$. The result is

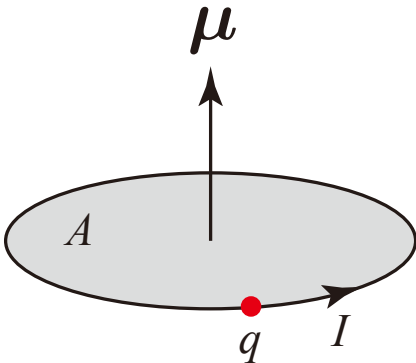
$$L = 1, 2, 3, 4, 5.$$

- Each angular momentum has z -components of. $m = -L, -L + 1, \dots, L, L + 1$.

[Magnetic moments]

- Relation between the magnetic moment and angular momentum**

Consider a charge q with mass m that orbits in a circle at radius r about the z -axis with angular velocity ω .



angular momentum

$$\begin{aligned}\mathbf{J} &= \mathbf{r} \times \mathbf{p} \\ &= r \times (mv)\hat{\mathbf{z}} \\ &= mr^2\omega\hat{\mathbf{z}}\end{aligned}$$

magnetic momentum

$$\mu = IA\hat{\mathbf{z}}$$

current $I = \frac{dq}{dt} = \frac{q}{P} = \frac{q\omega}{2\pi}$

area $A = \pi r^2$

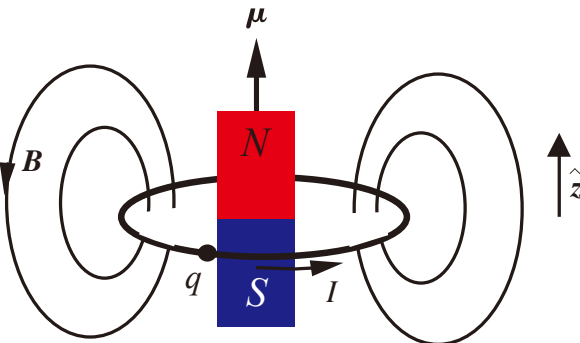
↑
period $P = \omega/2\pi$

}

$$\mu = \frac{1}{2}q\omega r^2\hat{\mathbf{z}}$$

Consequently, the classical relation between magnetic moment and angular momentum for an orbiting charge is

$$\mu = \frac{q}{2m}\mathbf{J}$$



Now consider a spinning sphere with uniform mass and charge distributions. It consists of many orbiting charge and mass elements, each obeying the above equation. Thus, the equation is also valid for the entire sphere.

In quantum mechanics, **the magnetic moment of the electron**, which is a point charge with zero radius, is given by

$$\mu_e = -g_e \frac{e}{2m_e} \mathbf{S} \quad (\text{where } g_e = 2.0 \text{ in Dirac equation})$$

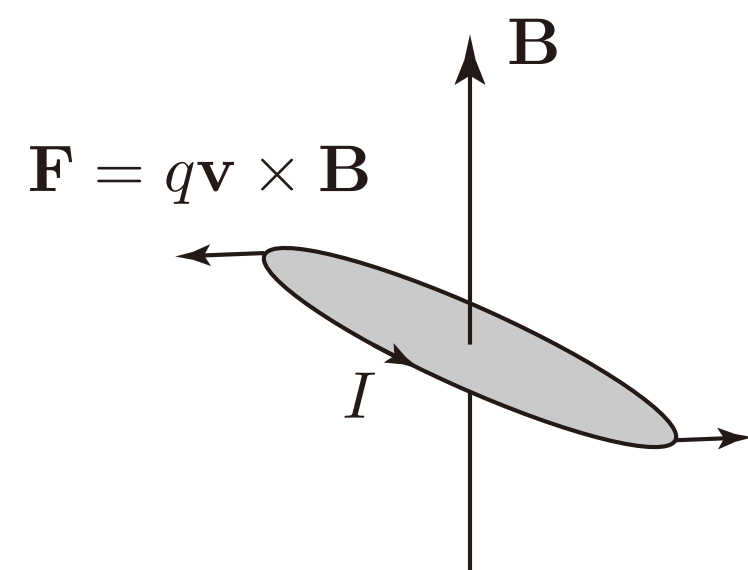
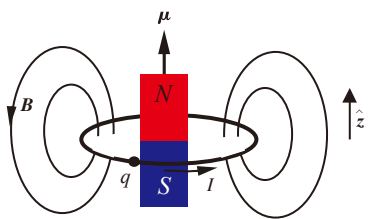
The magnetic moment of the electron is opposed to the direction of the spin $\mathbf{S}$, because of the negative charge.

Magnetic dipole in a magnetic field

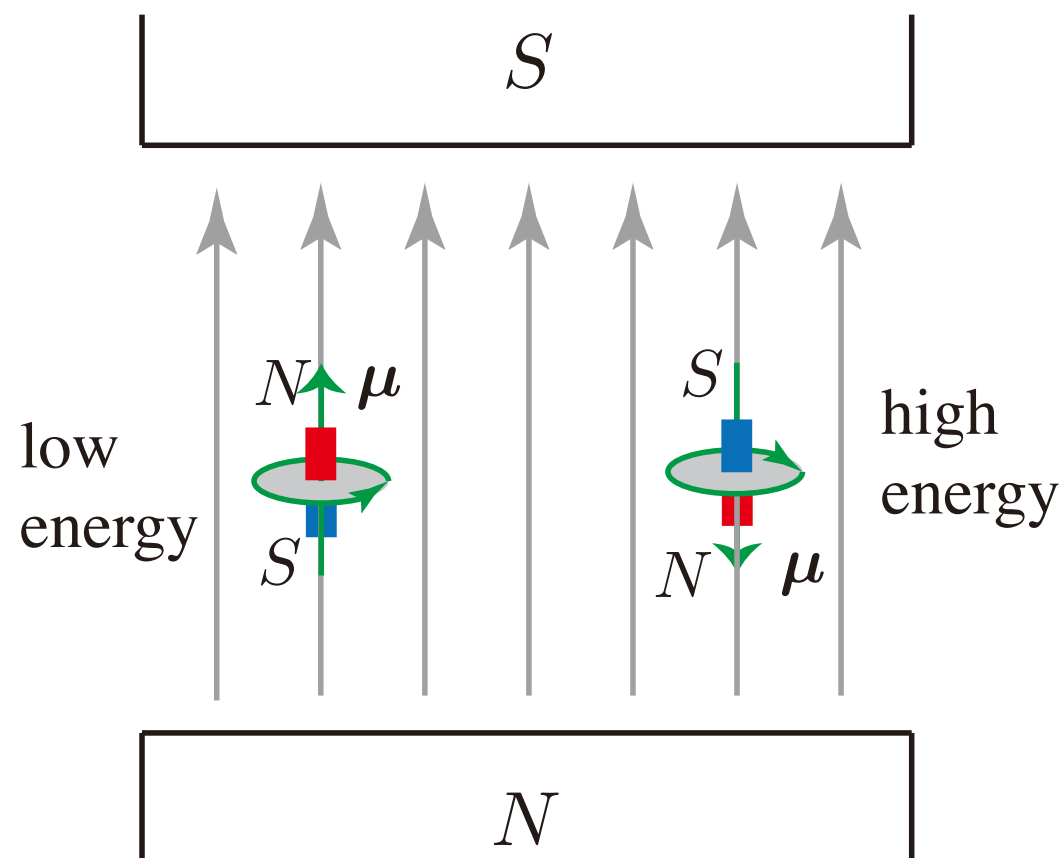
- A magnetic dipole will tend to align with an external magnetic field just as a compass needle aligns itself with the magnetic field of the earth.

One can visualize the external field acting on a magnet. This magnet experiences a torque tending to align it with the $\mathbf{B}$ field.

Integration of this torque over the appropriate angles yields a potential energy that is a function of the angle θ between the magnetic field $\mathbf{B}$ and magnetic moment $\boldsymbol{\mu}$ vector. The lowest potential energy occurs when the two vectors are aligned. The potential energy E_{pot} turns out to be the negative dot product of the vectors $\boldsymbol{\mu}$ and $\mathbf{B}$:



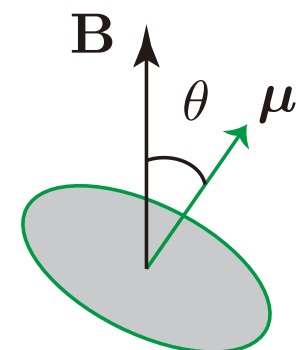
torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$



$$E_{\text{pot}} = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu B \cos \theta$$

$$E_{\text{pot}} < 0 \quad \text{if } \theta < \frac{\pi}{2}$$

$$E_{\text{pot}} > 0 \quad \text{if } \theta > \frac{\pi}{2}$$



$$E_{\text{pot}} = -\boldsymbol{\mu} \cdot \mathbf{B}$$

[Spin-Orbit Coupling]

- The fact that the remaining spin-orbit splitting is much smaller makes the LS coupling scheme a very useful one.
- **Fine-structure splitting:** Relativistic effects couple electron orbital angular momentum and electron spin to give the so-called fine structure in the energy levels. Inclusion of relativistic effects splits the **terms** into **levels** according to their J value.
- When the electron will move around the nucleus with a non relativistic velocity $\mathbf{v}$, the electric field exerting on the electron will be $\mathbf{E} = Ze\frac{\mathbf{r}}{r^3}$. (Note that the nucleus has a positive charge Ze .)

In the electron rest frame, this electric field will be perceived as a magnetic field

$$\begin{aligned}\mathbf{B}' &= \mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \boldsymbol{\beta} \times \mathbf{E}) = -\frac{\mathbf{v}}{c} \times \mathbf{E} \\ &= -\frac{Ze}{c} \frac{\mathbf{v} \times \mathbf{r}}{r^3} \\ &= \frac{Ze}{m_e c r^3} \mathbf{L} \quad (\mathbf{L} = \mathbf{r} \times \mathbf{p})\end{aligned}$$

Here, the magnetic field is perpendicular to the electron's orbital plane.

(where $\mathbf{L} \equiv \mathbf{r} \times \mathbf{p} = m_e \mathbf{r} \times \mathbf{v}$ is the electron's orbital angular momentum)

- This magnetic field will interact with the electron's **magnetic moment**, which is

$$\boldsymbol{\mu} = -\frac{e}{m_e c} \mathbf{S} = 2\boldsymbol{\mu}_{\text{classical}}$$

$$\text{definition of magnetic moment: } \boldsymbol{\mu} = \frac{q}{2m} \mathbf{J}$$

- Then, the interaction energy is

$$U = -\boldsymbol{\mu} \cdot \mathbf{B} = \frac{Ze^2}{m_e^2 c^2 r^3} \mathbf{S} \cdot \mathbf{L}$$

- For the sum of the interactions of all electrons will be

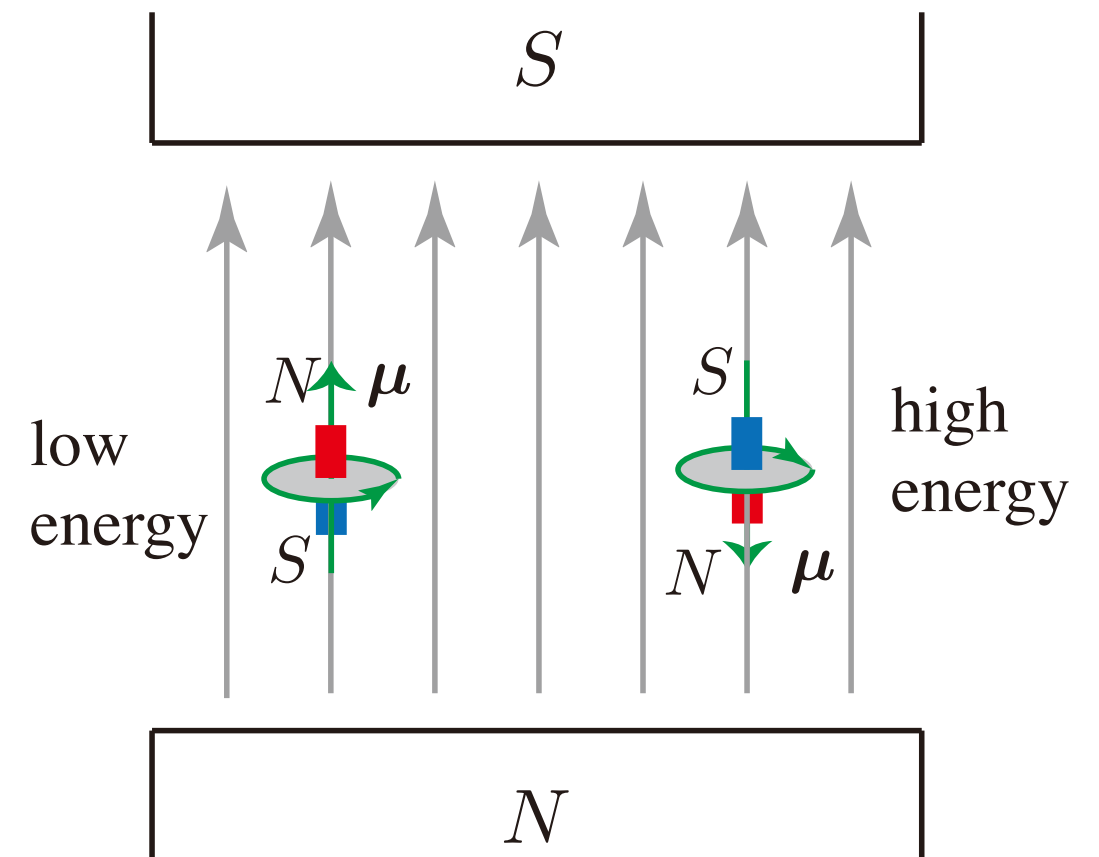
$$H_{\text{so}} = \xi (\mathbf{S} \cdot \mathbf{L})$$

From the relation $\mathbf{J}^2 = |\mathbf{L} + \mathbf{S}|^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{S} \cdot \mathbf{L}$,
we obtain

$$H_{\text{so}} = \frac{1}{2}\xi (\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2)$$

The eigen energy of the operator is given by

$$E_{\text{so}} = \frac{1}{2}C [J(J+1) - L(L+1) - S(S+1)]$$



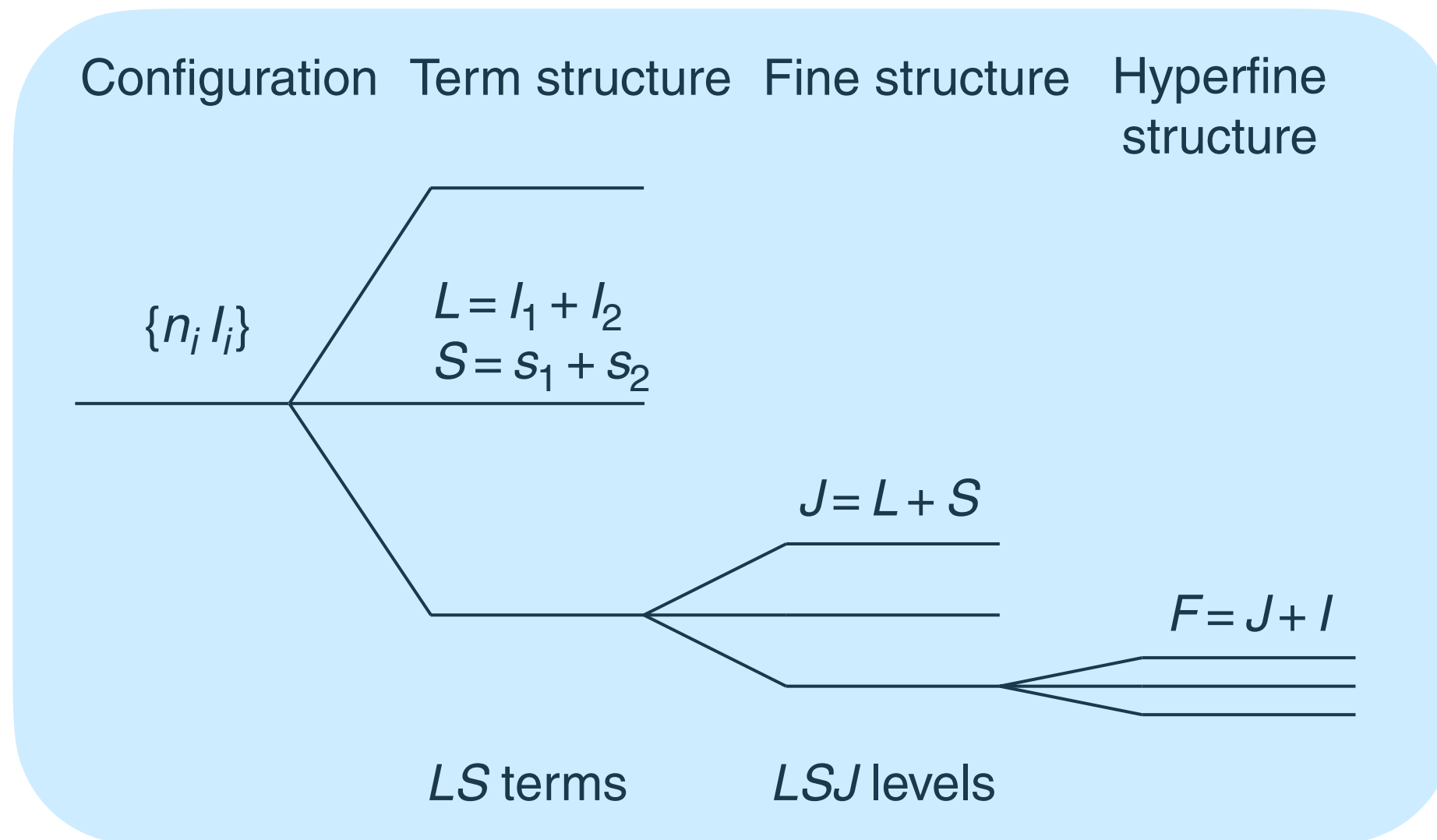
Note that

$$E_{\text{so}} = 0 \text{ for } L = 0 \text{ because } J = S$$

Therefore, there will be no fine structure for $L = 0$.

- Electronic configuration and energy level splitting

Configurations $\Rightarrow$ Terms $\Rightarrow$ Fine Structure (Spin-Orbit Interaction) $\Rightarrow$ Hyperfine Structure (Interaction with Nuclear Spin)



-
- **Active electrons:** As a result of the Pauli Principle, closed shells and sub-shells have both $L = 0$ and $S = 0$. This means that it is only necessary to consider ‘active’ electrons, those in open or partially-filled shells.

- **Parity of the wave function**

The parity of the wave function is determined by how the wave function behaves upon inversion. The square of the wave function, i.e., the probability distribution of the electrons, must be unchanged by the inversion operation.

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \pm \Psi(-\mathbf{r}_1, -\mathbf{r}_2, \dots, -\mathbf{r}_N)$$

Even parity states are given by + sign and odd parity states are given by - sign.

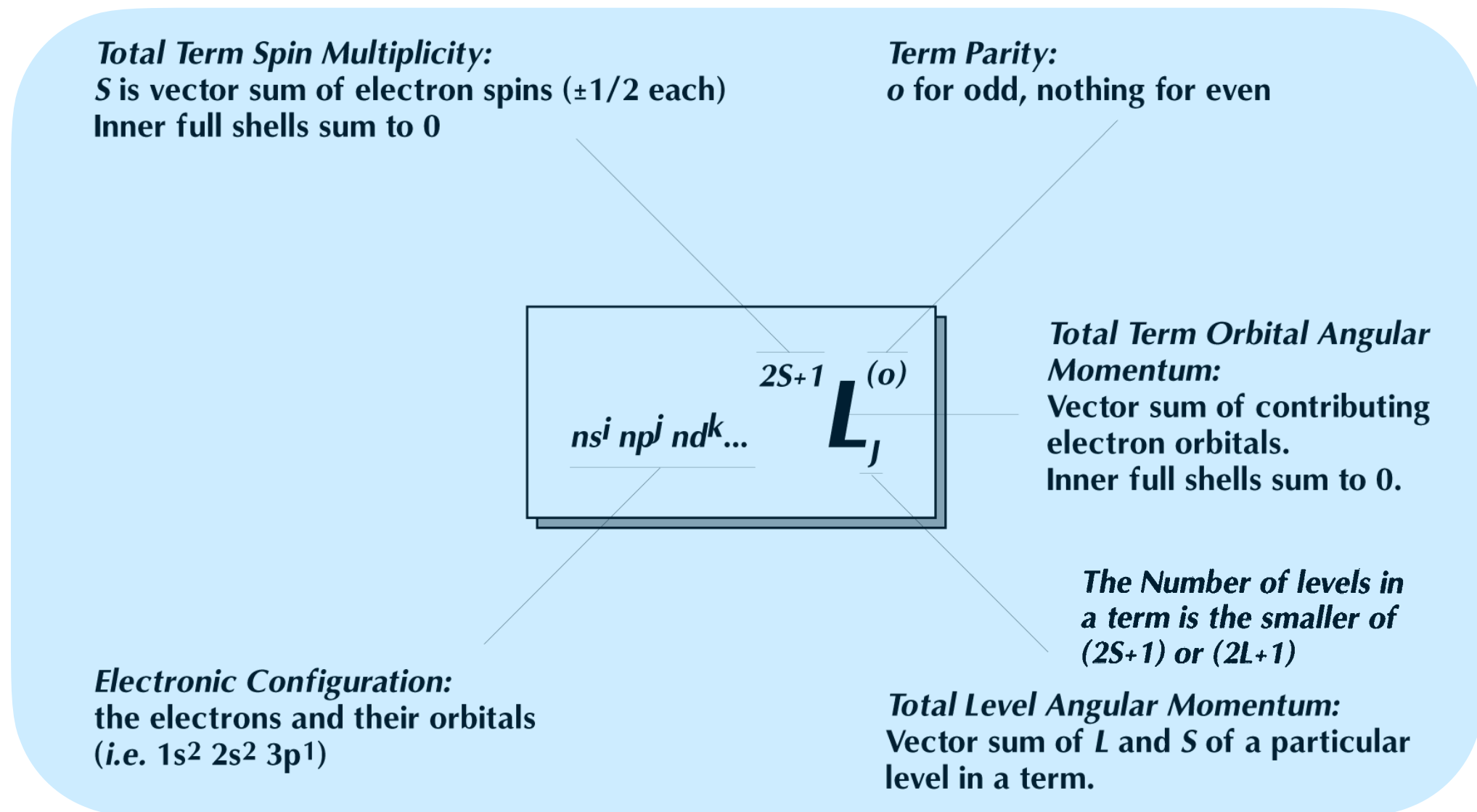
The parity arising from a particular configuration can be determined simply by summing the orbital angular momentum quantum numbers for each of the electrons.

$$\text{parity} \equiv (-1)^{l_1 + l_2 + \dots + l_N}$$

As closed shells and sub-shells have an even number of electrons, *it is only necessary to explicitly consider the active electrons.*

Spectroscopic Notation

- Spectroscopic Notation



- A state with $S = 0$ is a 'singlet' as $2S+1 = 1$.
 - ▶ $J = L$ (singlet)
- A state with $S = 1/2$ is a 'doublet' as $2S+1 = 2$
 - ▶ $J = L - 1/2, L + 1/2$ (doublet if $L \geq 1$)
- One with $S = 1$ is a 'triplet' as $2S+1 = 3$
 - ▶ $J = L - 1, L, L + 1$ (triplet $L \geq 1$)

$$n = 1, 2, 3, 4, 5 \dots \rightarrow K, L, M, N, O, \dots$$

$$\ell = 0, 1, 2, 3, 4 \dots \rightarrow s, p, d, f, g, \dots$$

$$L = 0, 1, 2, 3, 4 \dots \rightarrow S, P, D, F, G, \dots$$

sharp, principal, diffuse, fundamental,...

Energy ordering for Terms and Levels

- **Energy ordering: Hund's rules**

$$\text{Note that } H_{\text{so}} = \frac{1}{2}\xi (\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2)$$

(1) **S-rule:** For a given configuration, the state with the maximum spin multiplicity is lowest in energy.

The electrons repel each other, and therefore their mutual electrostatic energy is positive. The farther away the electrons get, the lower will be the contribution of the electrostatic energy to the total energy.

(2) **L-rule:** For a given configuration and spin multiplicity, the state with the maximum orbital angular momentum is the lowest in energy.

(3) **J-rule:** The lowest energy is obtained for lowest value of J in the normal case and for highest J value in the inverted case.

The normal case is a shell which is less than half filled. The inverted case is a shell which is more than half full such as the ground state of atomic oxygen.

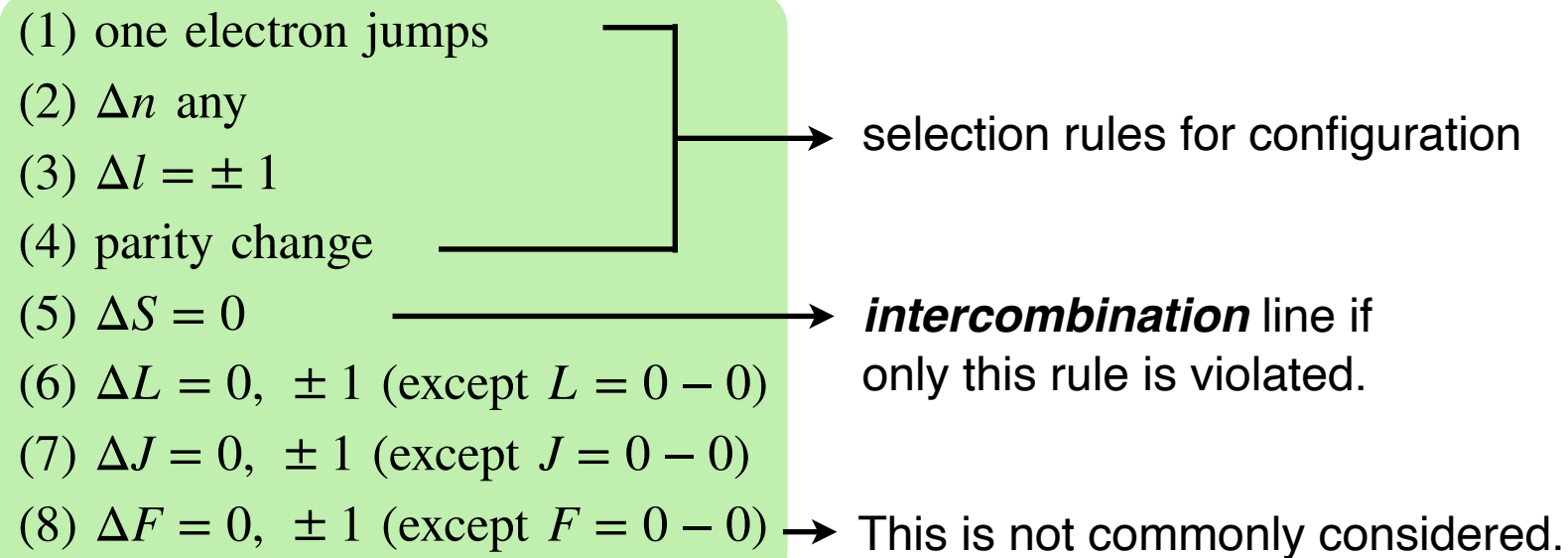
$${}^3P_0 < {}^3P_1 < {}^3P_2 \text{ for carbon } (1s^2 2s^2 2p^2)$$

$${}^3P_2 < {}^3P_1 < {}^3P_0 \text{ for oxygen } (1s^2 2s^2 2p^4)$$

- The Hund's rules are only applicable within LS coupling. They are only rigorous for ground states. However, they are almost always useful for determining the energy ordering of excited states. The rules show increasing deviations with higher nuclear charge.

Selection Rules

- **Selection Rules**

- 
- (1) one electron jumps
 (2) Δn any
 (3) $\Delta l = \pm 1$
 (4) parity change
- selection rules for configuration
- (5) $\Delta S = 0$
- **intercombination** line if only this rule is violated.
- (6) $\Delta L = 0, \pm 1$ (except $L = 0 - 0$)
 (7) $\Delta J = 0, \pm 1$ (except $J = 0 - 0$)
 (8) $\Delta F = 0, \pm 1$ (except $F = 0 - 0$)
- This is not commonly considered.

- **Allowed = Electric Dipole** : Transitions which satisfy all the above selection rules are referred to as **allowed transitions**. These transitions are strong and have a typical lifetime of $\sim 10^{-8}$ s. Allowed transitions are denoted without square brackets.

e.g., C IV 1548, 1550 Å

- Photons do not change spin, so transitions usually occur between terms with the same spin state ($\Delta S = 0$). However, relativistic effects mix spin states, particularly for high Z atoms and ions. As a result, one can get (weak) spin changing transitions. These are called **intercombination (semi-forbidden or intersystem) transitions** or lines. They have a typical lifetime of $\sim 10^{-3}$ s. An intercombination transition is denoted with a single right bracket.

C III] $2s^2 \ ^1S - 2s2p \ ^3P^o$ at 1908.7 Å. ($\Delta S = 1$)

- If any one of the rules 1-4, 6-8 are violated, they are called **forbidden transitions** or lines. They have a typical lifetime of $\sim 1 - 10^3$ s. A forbidden transition is denoted with two square brackets.

1906.7 Å [C III] $2s^2 \ ^1S_0 - 2s2p \ ^3P_2^o$, ($\Delta S = 1, \Delta J = 2$)

- **Resonance line** denotes the longest wavelength, dipole-allowed transition arising from the ground state of a particular atom or ion.

Forbidden Lines

- Forbidden lines are often difficult to study in the laboratory as collision-free conditions are needed to observe metastable states.
 - In this context, it must be remembered that laboratory ultrahigh vacuums are significantly denser than so-called dense interstellar molecular clouds.

Even in the best vacuum on Earth, frequent collisions knock the electrons out of these orbits (metastable states) before they have a chance to emit the forbidden lines.

- In astrophysics, low density environments are common. In these environments, the time between collisions is very long and an atom in an excited state has enough time to radiate even when it is metastable.

Forbidden lines of nitrogen ([N II] at 6548 and 6584 Å), sulfur ([S II] at 6716 and 6731 Å), and oxygen ([O II] at 3727 Å, and [O III] at 4959 and 5007 Å) are commonly observed in astrophysical plasmas. *These lines are important to the energy balance of planetary nebulae and H II regions.*

Since metastable states are rather common, forbidden transitions account for a significant percentage of the photons emitted by the ultra-low density gas in Universe.

Forbidden lines can account for up to 90% of the total visual brightness of objects such as emission nebulae.

- *The forbidden 21-cm hydrogen line is particularly important for radio astronomy as it allows very cold neutral hydrogen gas to be seen.*

Selection Rules are not applicable to collisional transitions

- Collisional transitions do not follow the selection rules of radiative transitions because they are not driven by the interaction of a photon with electric dipole of an atom or a molecule, but rather by short-range, intense interactions during physical contact.
 - **No “electric dipole” requirement:** Radiative selection rules exist because a photon has angular momentum (± 1) and is a weak perturbation. Collisional energy transfer involves the full, complex interaction potential between particles, which can exchange any amount of angular momentum.
 - **Strong, short-range potential:** Collisions are not weak perturbations; they involve direct contact (short-range) which can shake up the entire electron cloud, ignoring orbital symmetry (parity) that would otherwise prohibit a transition.
 - **Multipolar interactions:** Collisional processes can be driven by dipole, quadrupole or higher-order moments, or even through direct exchange of electrons. This allows for transitions that are forbidden under the dipole approximation ($\Delta l = \pm 1$).
 - **Parity Violations:** Collisions can easily induce transitions within the same parity because the incoming particle breaks the symmetry of the target atom/molecule.
 - It has been shown that rotational transitions can occur with changes in angular momentum much greater than the radiative selection rule ($\Delta J = \pm 1$), which some collisions causing changes of J by ten or more units.

[Notations]

- Notations for Spectral Emission Lines and for Ions

- There is a considerable confusion about the difference between these two ways of referring to a spectrum or ion, for example, C III or C⁺². These have very definite different physical meanings. However, in many cases, they are used interchangeably.

- **C⁺² is a baryon and C III is a set of photons.**

C⁺² refers to carbon with two electrons removed, so that is doubly ionized, with a net charge of +2.

- C III is the spectrum produced by carbon with two electrons removed.

C III will be produced **by impact excitation of C⁺² or by recombination of C⁺³.**

So, depending on how the spectrum is formed. **C III may be emitted by C⁺² or C⁺³.**

collisional excitation: $C^{+2} + e^{-} \rightarrow C^{+2*} + e^{-} \rightarrow C^{+2} + e^{-} + h\nu$

recombination: $C^{+3} + e^{-} \rightarrow C^{+2} + h\nu$

- There is no ambiguity in absorption line studies: **only C⁺² can produce a C III absorption line.** This had caused many people to think that C III refers to the matter rather than the spectrum.

But this notation is ambiguous in the case of emission lines.

[Hydrogen Atom] - Fine Structure

- Fine structure of the hydrogen atom

configuration	L	S	J	term	level
ns	0	$1/2$	$1/2$	2S	$^2S_{1/2}$
np	1	$1/2$	$1/2, 3/2$	$^2P^o$	$^2P_{1/2}^o, ^2P_{3/2}^o$
nd	2	$1/2$	$3/2, 5/2$	2D	$^2D_{3/2}, ^2D_{5/2}$
nf	3	$1/2$	$5/2, 7/2$	$^2F^o$	$^2F_{5/2}^o, ^2D_{7/2}^o$

Relativistic QM (Dirac's eq)

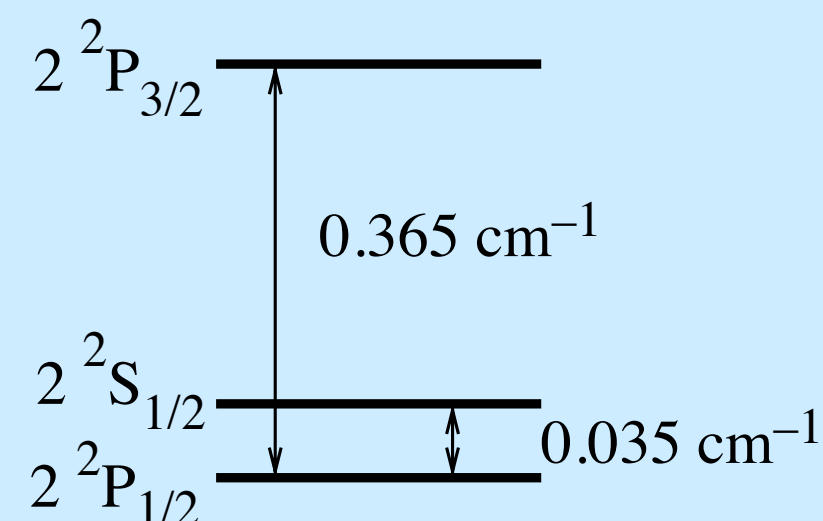
$$\begin{array}{rcl}
 & ^2P_{3/2} & \text{-----} \\
 & 2^2S_{1/2} = 2^2P_{1/2} & \text{-----} \\
 & & \\
 & 1^2S_{1/2} & \text{-----}
 \end{array}$$

- Splitting in the $n = 2$ levels of atomic hydrogen. The larger splitting is the fine structure and the smaller one the Lamb shift.

According to the Dirac equation, the $^2S_{1/2}$ and $^2P_{1/2}$ orbitals should have the same energies. However, the interaction between the electron and the vacuum (which is not accounted for by the Dirac equation) causes a tiny energy shift on $^2S_{1/2}$.

(Quantum electrodynamics effect)

Quantum Electrodynamics

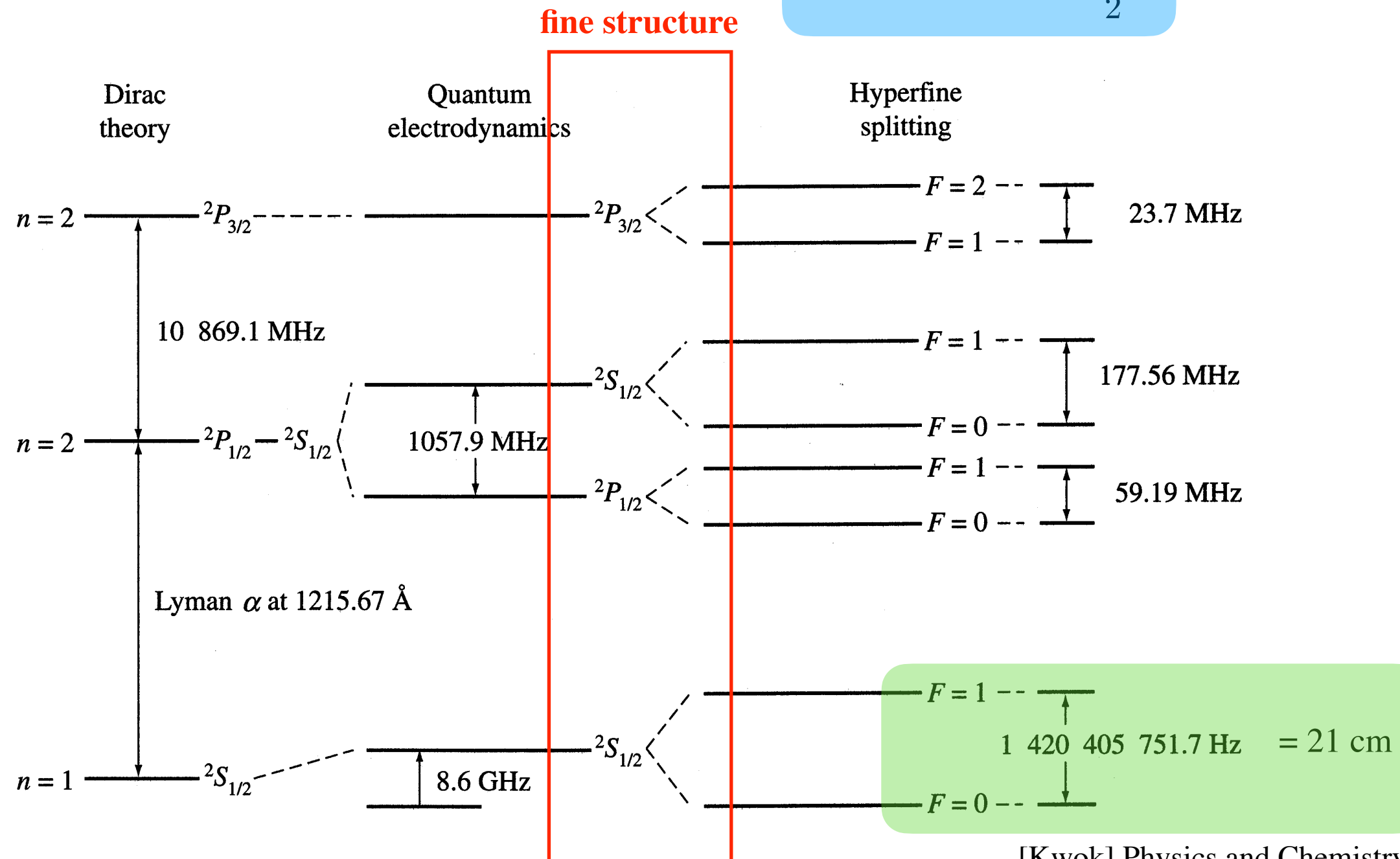


Hydrogen Atom : Fine Structure / Hyperfine Structure

- Hyperfine Structure in the H atom

Coupling the nuclear spin I to the total electron angular momentum J gives the final angular momentum F . For hydrogen this means

$$F = J + I = J \pm \frac{1}{2}$$



Hydrogen Atom : Allowed Transitions

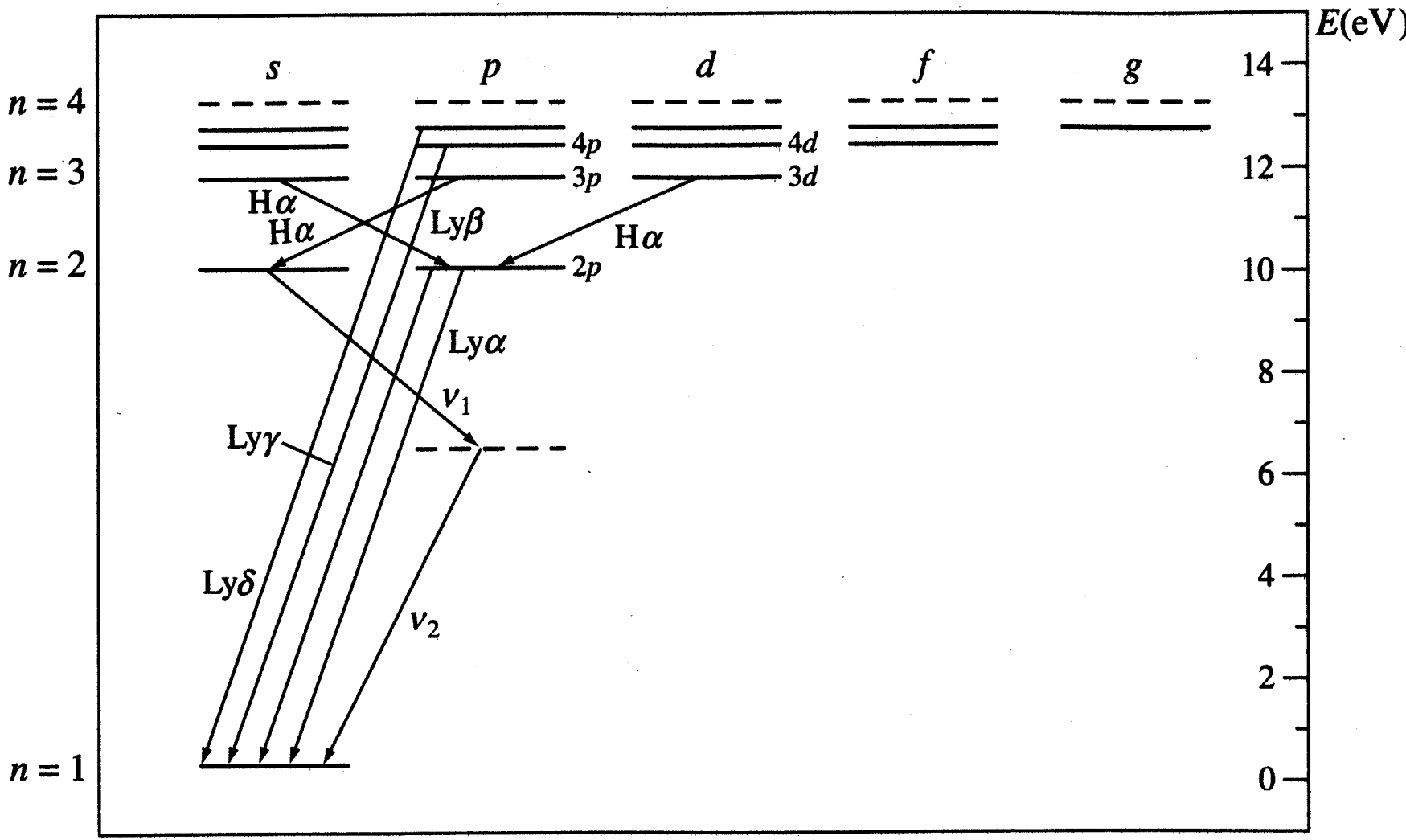
• Selection Rules

- Transitions are governed by selection rules which determine whether they can occur.

Δn any → selection rule for configuration
 $\Delta l = \pm 1$
 $\Delta S = 0$ → For H atom, this is always satisfied as $S = 1/2$ for all states.
 $\Delta L = 0, \pm 1$ (not $L = 0 - 0$)
 $\Delta J = 0, \pm 1$ (not $J = 0 - 0$)

For H-atom, l and L are equivalent since there is only one electron.

For H α transitions:

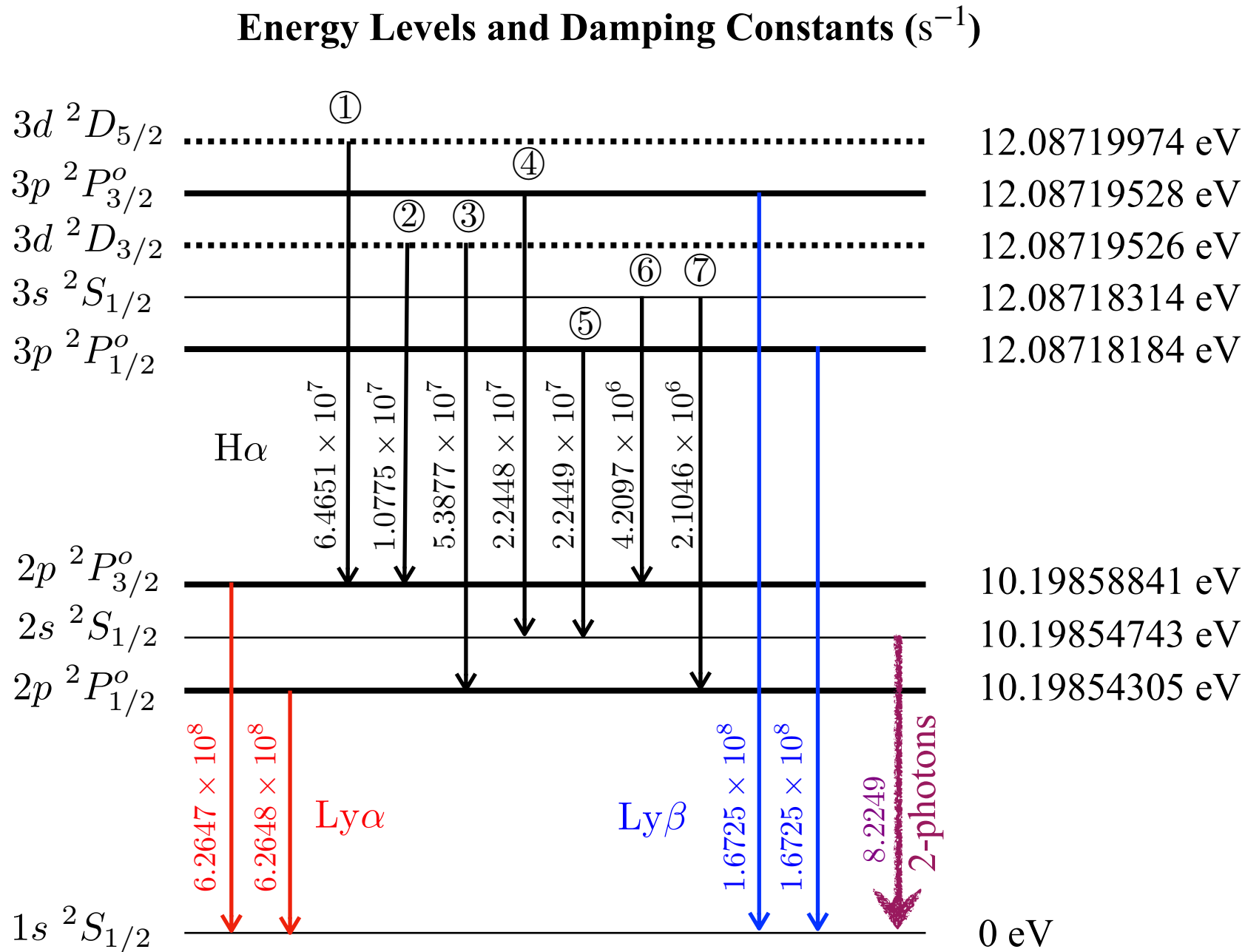


Not all H α transitions which correspond to $n = 2 - 3$ are allowed.

- $2s_{\frac{1}{2}} - 3p_{\frac{1}{2}}$ is allowed;
- $\quad \quad - 3p_{\frac{3}{2}}$ is allowed;
- $2p_{\frac{1}{2}} - 3d_{\frac{5}{2}}$ is not allowed; ($\Delta J = 2$)
- $\quad \quad - 3s_{\frac{1}{2}}$ is allowed;
- $\quad \quad - 3d_{\frac{3}{2}}$ is allowed;
- $2p_{\frac{3}{2}} - 3s_{\frac{1}{2}}$ is allowed;
- $\quad \quad - 3d_{\frac{3}{2}}$ is allowed;
- $\quad \quad - 3d_{\frac{5}{2}}$ is allowed.

The transition between $2s - 1s$ is not allowed ($\Delta l = 0$).

Hydrogen Atom: Allowed Transitions & Einstein A coefficients



Energy Levels of Neutral Hydrogen (NIST)

Configuration	Term	J	Level (cm ⁻¹)	Ref.
1s	² S	1/2	0.0000	MK00a
2p	² P ^o	1/2	82258.9191	MK00a
		3/2	82259.2850	MK00a
2s	² S	1/2	82258.9544	MK00a
3p	² P ^o	1/2	97492.2112	MK00a
		3/2	97492.3196	MK00a
3s	² S	1/2	97492.2217	MK00a
3d	² D	3/2	97492.3195	MK00a
		5/2	97492.3556	MK00a
4p	² P ^o	1/2	102823.8486	MK00a
		3/2	102823.8943	MK00a
4s	² S	1/2	102823.8530	MK00a
4d	² D	3/2	102823.8942	MK00a
		5/2	102823.9095	MK00a
4f	² F ^o	5/2	102823.9095	MK00a
		7/2	102823.9171	MK00a
5p	² P ^o	1/2	105291.6287	MK00a
		3/2	105291.6521	MK00a
5s	² S	1/2	105291.6309	MK00a
5d	² D	3/2	105291.6520	MK00a
		5/2	105291.6599	MK00a
5f	² F ^o	5/2	105291.6598	MK00a
		7/2	105291.6637	MK00a
5g	² G	7/2	105291.6637	MK00a
		9/2	105291.6661	MK00a
H	<i>Limit</i>		109678.7717	MK00a

• Hydrogen: *lifetime of excited states*

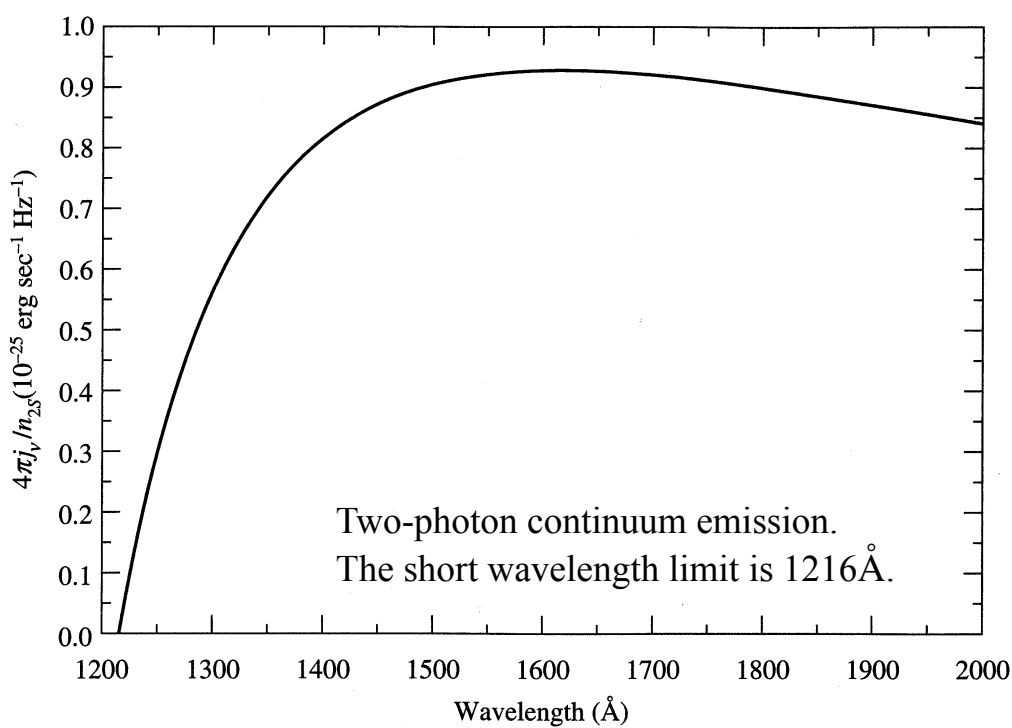
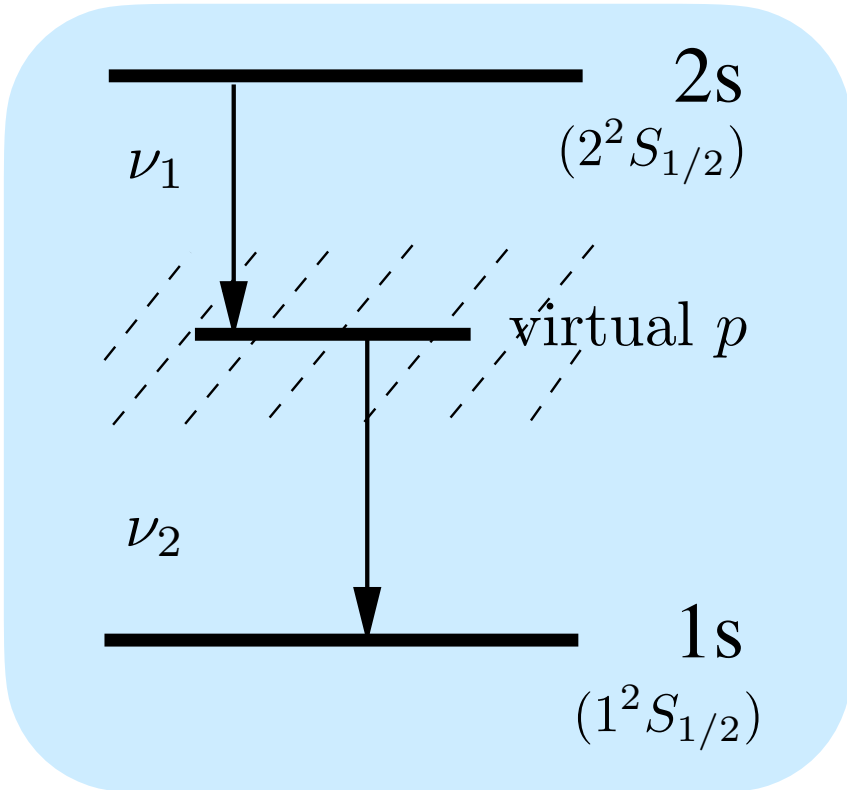
$$\tau_i = \left(\sum_j A_{ij} \right)^{-1} \quad \text{where } A_{ij} \text{ is the Einstein A coefficient}$$

Level	2s	2p	3s	3p	3d
τ/s	0.14	1.6×10^{-9}	1.6×10^{-7}	5.4×10^{-9}	2.3×10^{-7}

- Lifetimes for allowed transitions are short, a few times 10^{-9} s.
- However, the lifetime for the (2s) $2^2S_{1/2}$ level is ~ 0.14 s, which is 10^8 times longer than the 2p states. (*The level is called to be metastable.*)

• *Two-photon continuum radiation*

- In low-density environments (e.g., ISM), an electron in the $2^2S_{1/2}$ level can jumps to a virtual p state, which lies between $n = 1$ and $n = 2$ levels. The electron then jumps from this virtual state to the ground state, in the process emitting two photons with total frequency $\nu_1 + \nu_2 = \nu_{\text{Ly}\alpha}$.
- Since this virtual p state can occur anywhere between $n = 1$ and $n = 2$, continuum emission longward of Ly α will result.
- Because the radiative lifetime of the 2s level is long. we need to consider the possibility for collisions with electrons and protons to depopulate 2s level before a spontaneous decay occurs.
- However, *the critical density, at which deexcitation by electron and proton collision is equal to the radiative decay rate, is $n_{\text{crit}} \approx 1880 \text{ cm}^{-3}$.* **In the ISM, the radiative decay is in general faster than the collisional depopulation process.**



[Helium Atom]

(1) The ground state is $1s^2$.

This is a closed shell, with $L = 0$ and $S = 0$, hence it gives rise to a single, even parity term 1S , and level 1S_0 .

S	L	J
0	0	0

 $^1S \rightarrow ^1S_0$

(2) The first excited configuration is $1s2s$.

This has $l_1 = l_2 = 0$ and hence $L = 0$, but $s_1 = s_2 = \frac{1}{2}$ giving both $S = 0$ (singlet) or $S = 1$ (triplet) states.

S	L	J
0	0	0
1	0	1

 $^1S \rightarrow ^1S_0$
 $^3S \rightarrow ^3S_1$

For a given configuration, the state with the maximum spin multiplicity is lowest in energy.

So the 3S term (3S_1 level) is lower in energy than the 1S term (1S_0 level). In practice, the splitting between these terms is 0.80 eV.

(3) The next excited configuration is $1s2p$, which has odd parity.

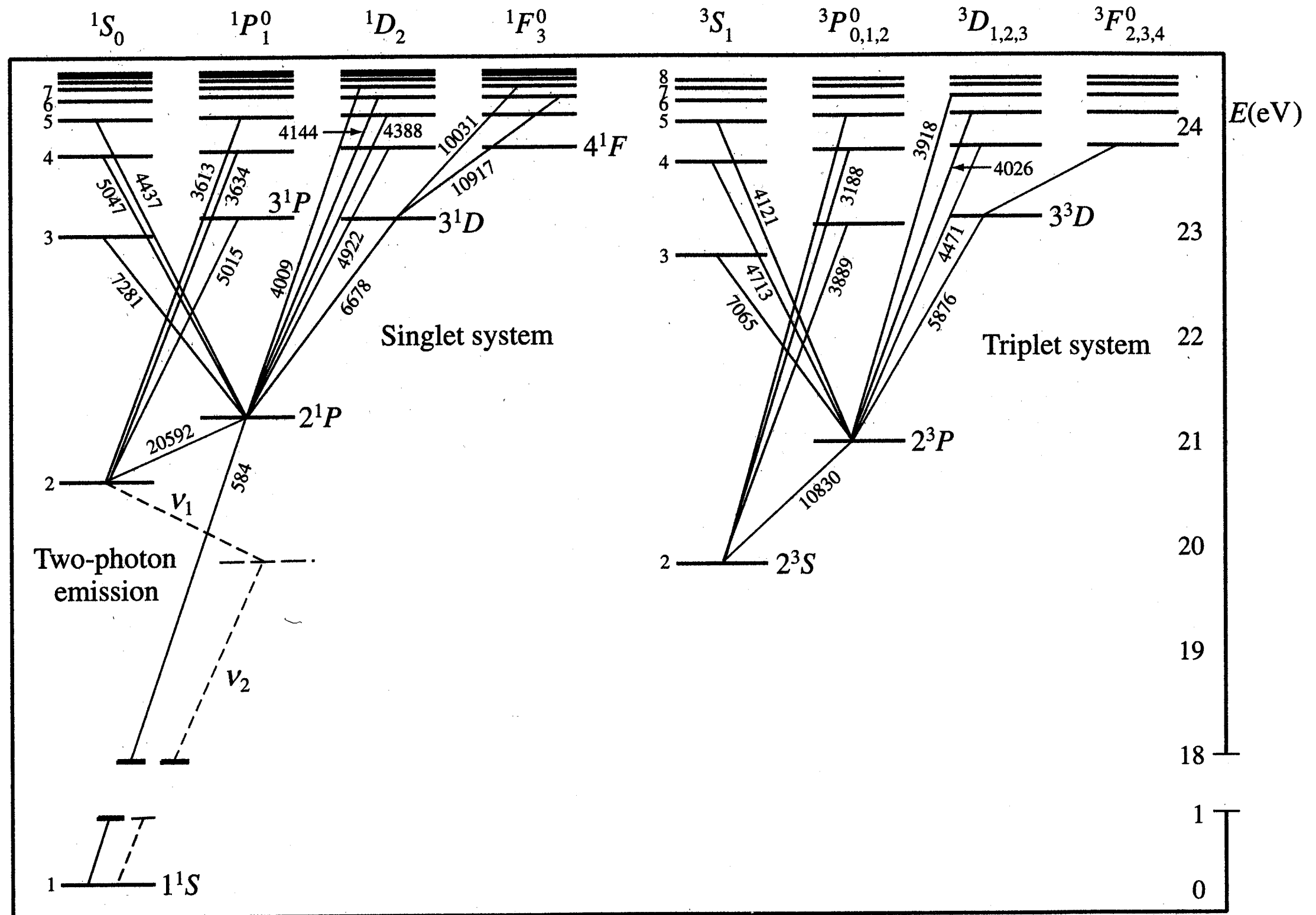
This has $l_1 = 0$ and $l_2 = 1$, giving $L = 1$; again $s_1 = s_2 = \frac{1}{2}$, giving both $S = 0$ and $S = 1$ terms.

S	L	J
0	1	1
1	1	0, 1, 2

 $^1P^o \rightarrow ^1P_1^o$
 $^3P^o \rightarrow ^3P_0^o < ^3P_1^o < ^3P_2^o$

Following the rule above, the $^3P^o$ term is lower in energy than the $^1P^o$ term, in this case by 0.25 eV.

- Helium (Grotrian diagram)



The states can be divided into two separate groups because of the selection rule $\Delta S = 0$.

[LS Terms] - (1) Nonequivalent Electrons, 2p3p

- **Equivalent and Nonequivalent Electrons**

Nonequivalent electrons are those differing in either n or l values, whereas equivalent electrons have the same n and l values.

- **Consider the combination of two p electrons.**

(1) **2p3p** - Two electrons are nonequivalent.

In this nonequivalent case, all possible spectroscopic combinations are available.

$$S = 0, 1, \quad L = 0, 1, 2 \quad \rightarrow \quad {}^1S, {}^1P, {}^1D, {}^3S, {}^3P, {}^3D$$

$\rightarrow$ 6 spectroscopic terms

$$\rightarrow \quad {}^1S_0, {}^1P_1, {}^1D_2, {}^3S_1, {}^3P_{0,1,2}, {}^3D_{1,2,3}$$

$\rightarrow$ 10 spectroscopic levels

We can have $(2J+1)$ states for each J value.

Therefore, 36 distinguishable states are available in total.

$$\rightarrow \quad 1 + 3 + 5 + 3 + 9 + 15 = 36$$

There are two possible states $m_s = \pm 1/2$ for spin of each electron, and three states $m_l = -1, 0, 1$ for orbital angular momentum of each electron.

Thus, we expect that there will be $2^2 \times 3^2 = 36$ distinguishable states.

(2) Equivalent Electrons, $2p^2$

(2) $2p^2$ - Two electrons are equivalent.

Then, *all the 36 states are not available. Some are ruled out by Pauli exclusion principle, and some are ruled out because they are not distinguishable from others.*

How to make spectroscopic terms

- The first step is to make a table to label the states for a single electron (e.g., a, b, c, d, e, f), as shown in Table 1.

Table 1

Label	m_l	m_s
a	+1	+1/2
b	0	+1/2
c	-1	+1/2
d	+1	-1/2
e	0	-1/2
f	-1	-1/2

Table 2

	States	M_L	M_S
1	ab	1	1
2	ac	0	1
3	ad	2	0
4	ae	1	0
5	af	0	0
6	bc	-1	1
7	bd	1	0
8	be	0	0
9	bf	-1	-1
10	cd	0	0
11	ce	-1	0
12	cf	-2	0
13	de	1	-1
14	df	0	-1
15	ef	-1	-1

- Next step is to make a table for the combination of (M_L, M_S) of two electrons, as shown in Table 2. Here, $M_L = m_{l1} + m_{l2}$, and $M_S = m_{s1} + m_{s2}$.
 - According to Pauli’s exclusion principle, the states that have two identical states (aa, bb, cc, dd, ee, and ff) are not allowed.
 - Notice also that “ab” and “ba” states are identical and thus the “ba” state is ignored. Other identical combinations are also ignored.
- Following the above two rules, we construct Table 2 which contains 15 distinguishable states.

Table 3

States	M_L	M_S	Term 1	Term 2	Term 3
ab	1	1		3P	
ac	0	1		3P	
ad	2	0	1D		
ae	1	0	1D		
af	0	0	1D		
bd	1	0		3P	
be	0	0		3P	
cd	0	0			1S

- In addition to the above two rules, we can recognize that every “negative” values have always their “positive” counterparts.
 - Therefore, it is more convenient to remove the states with negative values. This gives us Table 3, which contains only 8 states.
- Now, we pick the states **starting with the largest M_L and then the largest M_S** .
 - (ad) $M_L = 2$ and $M_S = 0$: The presence of the $M_L = 2, M_S = 0$ indicates that a 1D term is among the possible terms. To this term we must further assign states with $M_L = 1, 0$ and $M_S = 0$ (ae, af). What is left?
 - (ab) $M_L = 1$ and $M_S = 1$: This is the next largest values. The combination $M_L = 1, M_S = 1$ indicates that a 3P term is among the possible terms. To this term we must further assign states with $M_L = 1, 0$ and $M_S = 1, 0$ (ac, bd, be). What is left?
 - (cd) $M_L = 0$ and $M_S = 0$: This is the only remaining combination. This implies that a 1S term is among the possible terms.
 - **Finally, we obtain 3 terms $^1D, ^3P$, and 1S .**
 - **The 3 terms are split into 5 levels : $^1D_2, ^3P_{0,1,2}$, and 1S_0 .**

(3) Another method for $2p^2$

When we have only two electrons, we can use the Pauli principle to obtain the terms.

This method is much simpler than the above method. However, this method is not easy to apply to the case of three electrons.

Recall that the Pauli principle states that **the total eigenfunction must be antisymmetric with respect to the exchange of two particles**. Therefore, we can have only two distinct cases:

- (a) symmetric function for the spin + antisymmetric function for the orbital angular momentum $\Rightarrow {}^3P$
- (b) antisymmetric function for the spin + symmetric function for the orbital angular momentum $\Rightarrow {}^1S, {}^1D$

Note that among the six terms ${}^1S, {}^1P, {}^1D, {}^3S, {}^3P, {}^3D$, the following terms are all excluded.

1P is antisymmetric for both spin and orbital angular momenta

3S is symmetric for both spin and orbital angular momenta

3D is also symmetric for both spin and orbital angular momenta

Because

- $S = 0 \rightarrow s_1 = 1/2, s_2 = -1/2$: Product of two spin functions are antisymmetric w.r.t. the exchange
- $S = 1 \rightarrow s_1 = 1/2, s_2 = 1/2$: Product of two spin functions are symmetric.
- $L = 1 \rightarrow l_1 = 1, l_2 = 0$: The first wavefunction is antisymmetric and the second one is symmetric.
Therefore, their product is antisymmetric w.r.t. the exchange.
- $L = 2 \rightarrow l_1 = 1, l_2 = 1$: Both are antisymmetric. Therefore, their product is symmetric.

(4) Equivalent Electrons, 2p³

(4) **2p³** - Three electrons are equivalent.

- According to Pauli’s exclusion principle, any states that include two identical states (aaa, aab, aac, add, bbc, bbd, etc) are not allowed.
- We have only seven states that have non-negative values, as shown in Table 4.

Table 4

States	M _L	M _S	Term 1	Term 2	Term 3
abc	0	3/2			⁴ S
abd	2	1/2	² D		
abe	1	1/2	² D		
abf	0	1/2	² D		
acd	1	1/2		² P	
ace	0	1/2		² P	
bcd	0	1/2			⁴ S

Table 1

Label	m _l	m _s
a	+1	+1/2
b	0	+1/2
c	-1	+1/2
d	+1	-1/2
e	0	-1/2
f	-1	-1/2

- Now, we pick the states starting with the largest M_L and then the largest M_S .
 - (abd) $M_L = 2$ and $M_S = 1/2$: This indicates that a ²D term is among the possible terms. To this term we must further assign states with $M_L = 1,0$ and $M_S = 1/2$ (abe, abf). What is left?
 - (acd) $M_L = 1$ and $M_S = 1/2$: This indicates the presence of a ²P term. To this term we must further assign states with $M_L = 0$ and $M_S = 1/2$ (ace). What is left?
 - (abc) $M_L = 0$ and $M_S = 3/2$: This indicates the presence of a ⁴S term.
 - *Finally, we obtain three terms (²D, ²P, and ⁴S) and five levels : ²D_{3/2,5/2}, ²P_{1/2,3/2}, and ⁴S_{3/2}.*

(5) Example : Doubly Ionized Oxygen, O III

Consider O III with the configuration: $1s^2 2s^2 2p 3d$.

$1s^2$ and $2s^2$ are closed, so they contribute no angular momentum.

For the 2p electron $l_1 = 1$ and $s_1 = \frac{1}{2}$;

for the 3d electron $l_2 = 2$ and $s_2 = \frac{1}{2}$.

$$\underline{L} = \underline{l}_1 + \underline{l}_2 \Rightarrow L = 1, 2, 3;$$

$$\underline{S} = \underline{s}_1 + \underline{s}_2 \Rightarrow S = 0, 1.$$

Combining these using all possible combinations of L and S , and the rules of vector addition, gives:

	L	S	J	Level
$\underline{J} = \underline{L} + \underline{S} \Rightarrow$	1	0	1	$^1P_1^o$
	1	1	0, 1, 2	$^3P_0^o, ^3P_1^o, ^3P_2^o$
	2	0	2	$^1D_2^o$
	2	1	1, 2, 3	$^3D_1^o, ^3D_2^o, ^3D_3^o$
	3	0	3	$^1F_3^o$
	3	1	2, 3, 4	$^3F_2^o, ^3F_3^o, ^3F_4^o$.

In total, 6 terms and 12 levels.

(6) Example: 4p4d

(Example) 4p4d electron configuration

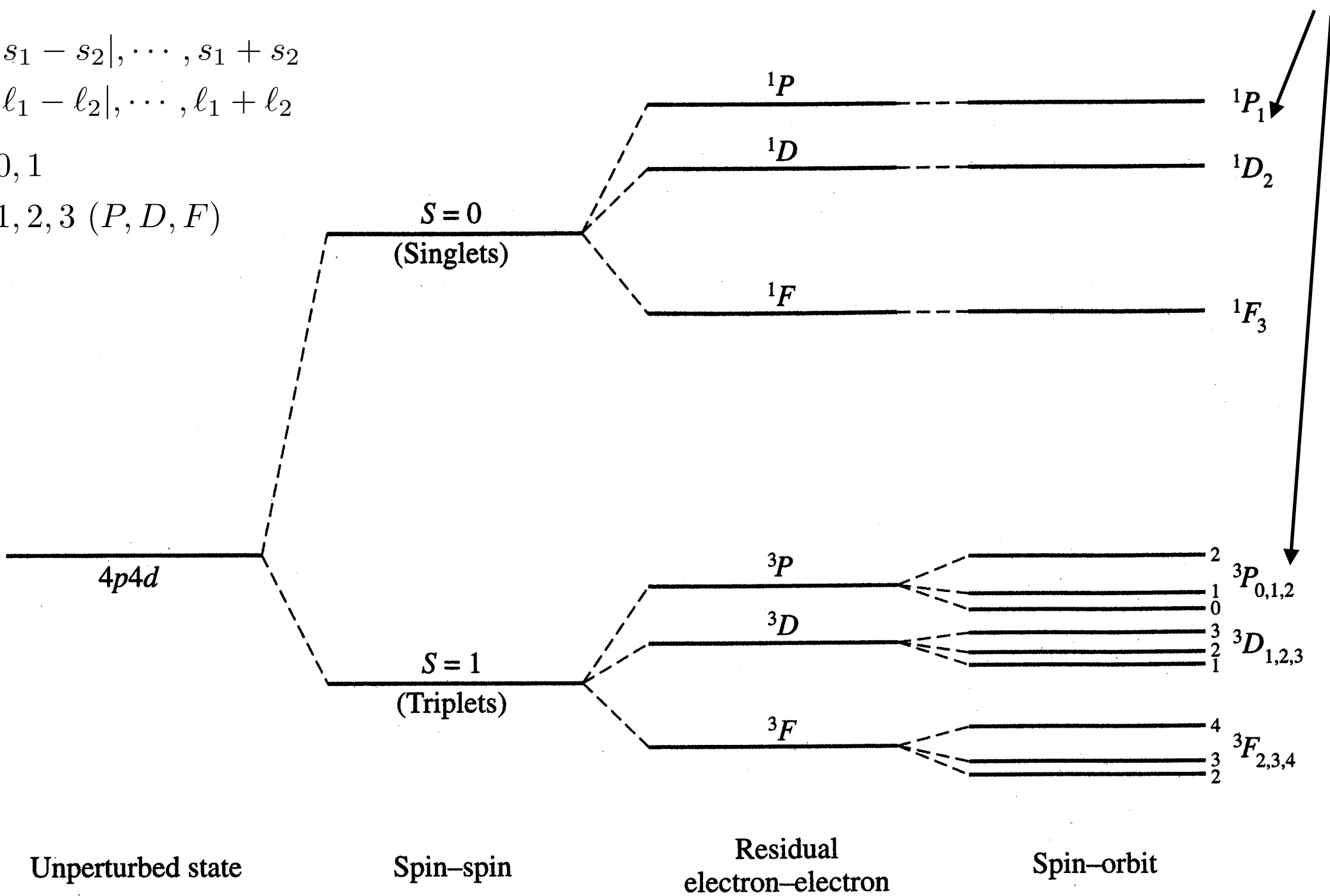
$S = |s_1 - s_2|, \dots, s_1 + s_2$

$L = |\ell_1 - \ell_2|, \dots, \ell_1 + \ell_2$

$S = 0, 1$

$L = 1, 2, 3 \text{ (P, D, F)}$

$J = |L - S|, \dots, L + S$



(7) Example np n'p

(Example) $npn'p$ electron configuration

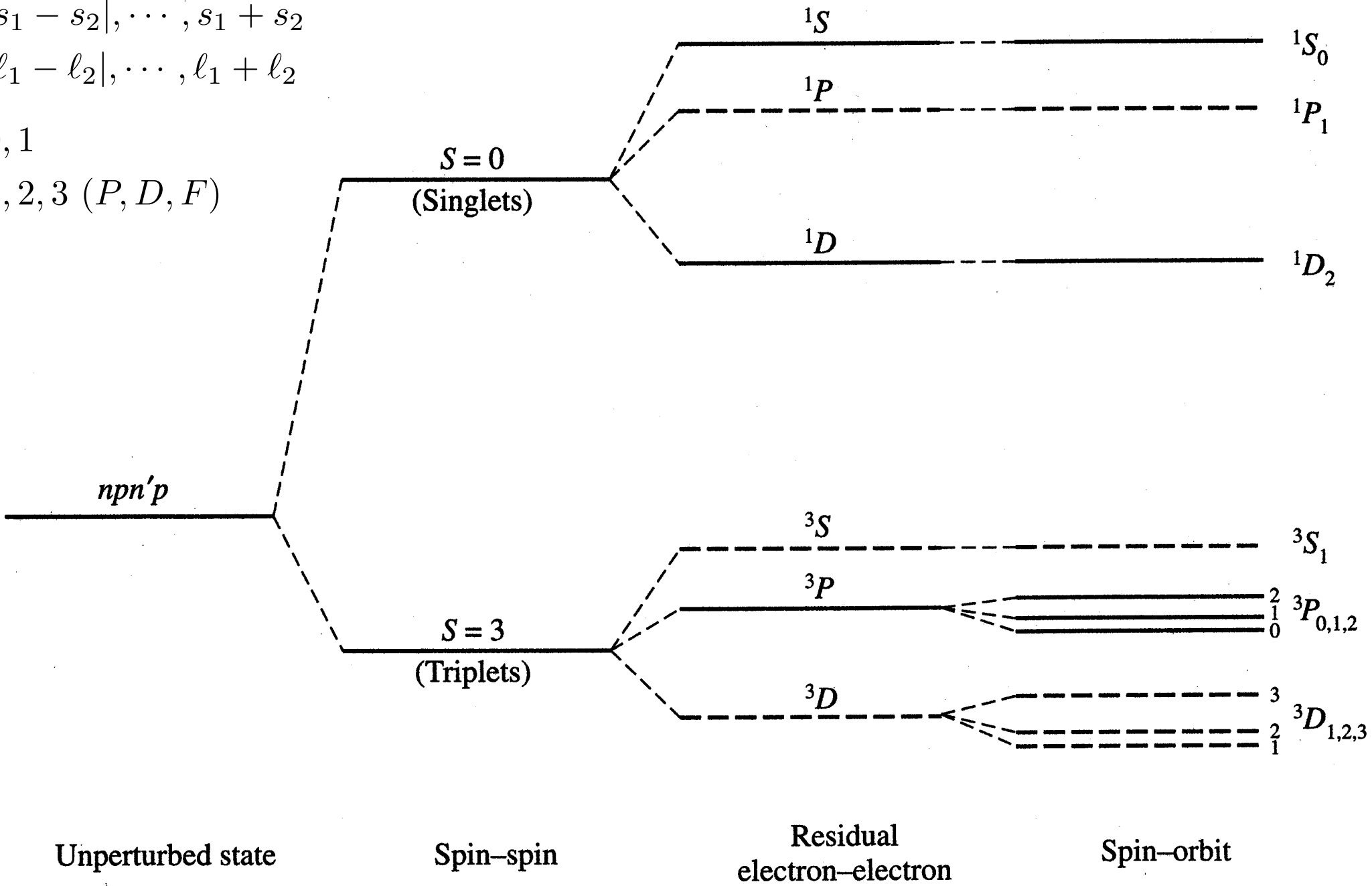
The dashed levels are missing if the two electrons are equivalent ($n = n'$)

$S = |s_1 - s_2|, \dots, s_1 + s_2$

$L = |\ell_1 - \ell_2|, \dots, \ell_1 + \ell_2$

$S = 0, 1$

$L = 1, 2, 3 \text{ (} P, D, F \text{)}$



(8) Terms for Ground configuration

[Kowk, Physics and Chemistry of the ISM]

Atomic terms arising from p^n and d^n configurations

Electron configuration	Terms
p^1, p^5	2P
p^2, p^4	$^1S, ^1D, ^3P$
p^3	$^2P, ^2D, ^4S$
d^1, d^9	2D
d^2, d^8	$^1S, ^1D, ^1G, ^3P, ^3F$
d^3, d^7	$^2P, ^2D, ^2F, ^2G, ^2H, ^4P, ^4F$
d^4, d^6	$^1S, ^1D, ^1F, ^1G, ^1I, ^3P, ^3D, ^3F, ^3G, ^3H, ^5D$
d^5	$^2S, ^2P, ^2D, ^2F, ^2G, ^2H, ^2I, ^4P, ^4D, ^4F, ^4G, ^6S$

Source: Bernath 1995, *Spectra of Atoms and Molecules*, Table 5.6.

[Draine, Physics of the ISM and IGM]

Table 4.1 Terms for ns and np Subshells

Ground configuration	Terms (in order of increasing energy)	Examples
$\dots ns^1$	$^2S_{1/2}$	H I, He II, C IV, N V, O VI
$\dots ns^2$	1S_0	He I, C III, N IV, O V
$\dots np^1$	$^2P^{\circ}_{1/2,3/2}$	C II, N III, O IV
$\dots np^2$	$^3P_{0,1,2}, ^1D_2, ^1S_0$	C I, N II, O III, Ne V, S III
$\dots np^3$	$^4S^{\circ}_{3/2}, ^2D^{\circ}_{3/2,5/2}, ^2P^{\circ}_{1/2,3/2}$	N I, O II, Ne IV, S II, Ar IV
$\dots np^4$	$^3P_{2,1,0}, ^1D_2, ^1S_0$	O I, Ne III, Mg V, Ar III
$\dots np^5$	$^2P^{\circ}_{3/2,1/2}$	Ne II, Na III, Mg IV, Ar IV
$\dots np^6$	1S_0	Ne I, Na II, Mg III, Ar III

Atom	K	L		M			N			O	Ground level
	1s	2s	2p	3s	3p	3d	4s	4p	4d	5s	
H	1	1									$^2S_{1/2}$
He	2	2									1S_0
Li	3	2	1								$^2S_{1/2}$
Be	4	2	2								1S_0
B	5	2	2	1							$^2P^{\circ}_{1/2}$
C	6	2	2	2							3P_0
N	7	2	2	3							$^4S^{\circ}_{3/2}$
O	8	2	2	4							3P_1
F	9	2	2	5							$^2P^{\circ}_{3/2}$
Ne	10	2	2	6							1S_0
Na	11	2	2	6	1						$^2S_{1/2}$
Mg	12				2						1S_0
Al	13				2	1					$^2P^{\circ}_{1/2}$
Si	14				2	2					3P_0
P	15				2	3					$^4S^{\circ}_{3/2}$
S	16				2	4					3P_2
Cl	17				2	5					$^2P^{\circ}_{3/2}$
Ar	18				2	6					1S_0
K	19	2	2	6	2	6	1				$^2S_{1/2}$
Ca	20						2				1S_0
Se	21					1	2				$^2D_{3/2}$
Ti	22					2	2				3F_2
V	23					3	2				$^4F_{3/2}$
Cr	24					5	1				7S_3
Mn	25					5	2				$^6S_{5/2}$
Fe	26					6	2				5D_4
Co	27					7	2				$^4F_{9/2}$
Ni	28					8	2				3F_4
Cu	29	2	2	6	2	6	10	1			$^2S_{1/2}$
Zn	30							2			1S_0

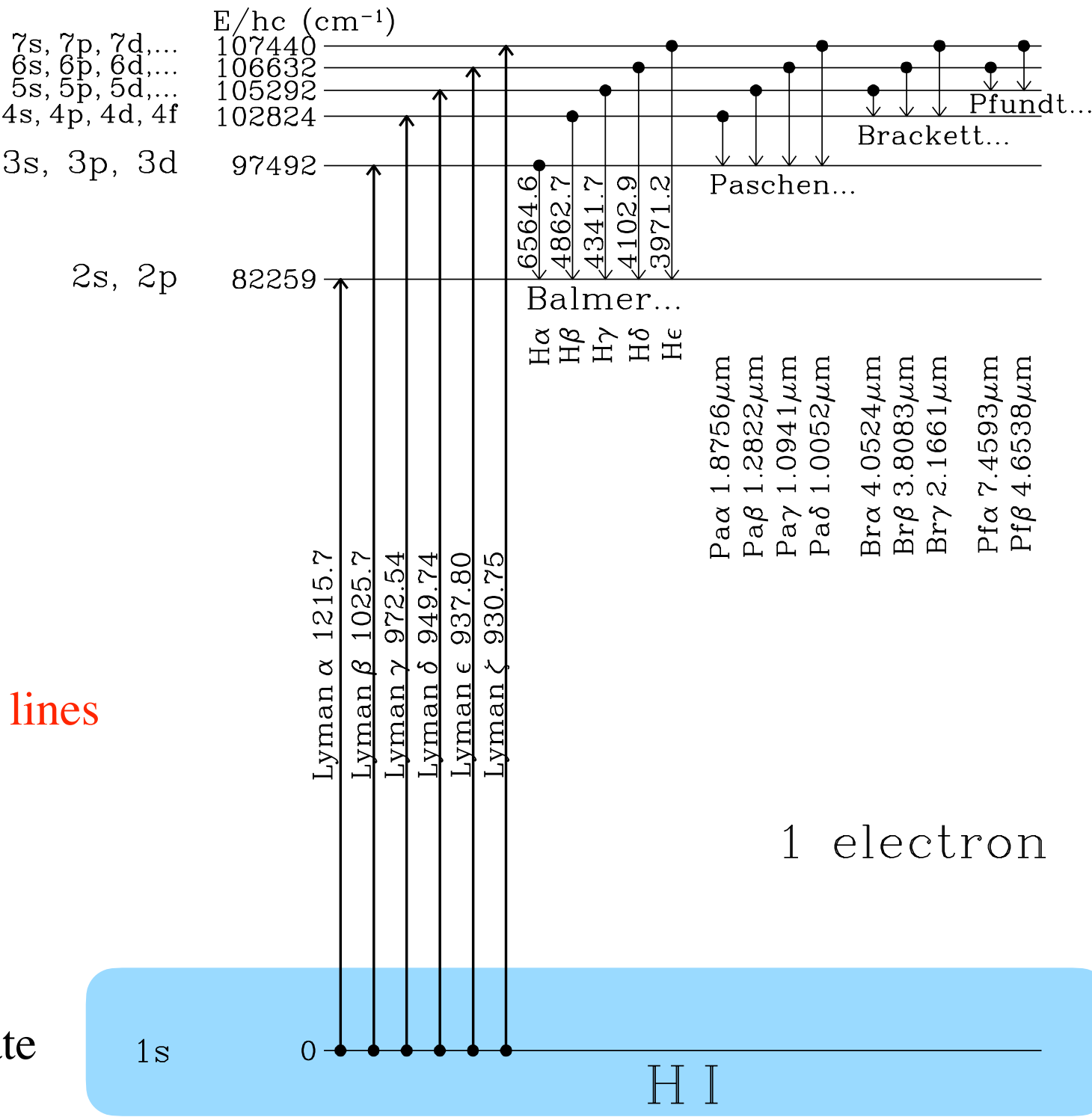
Blue: No fine structure in the ground state.

[Energy Level Diagrams]

- 1 electron

resonance lines

ground state



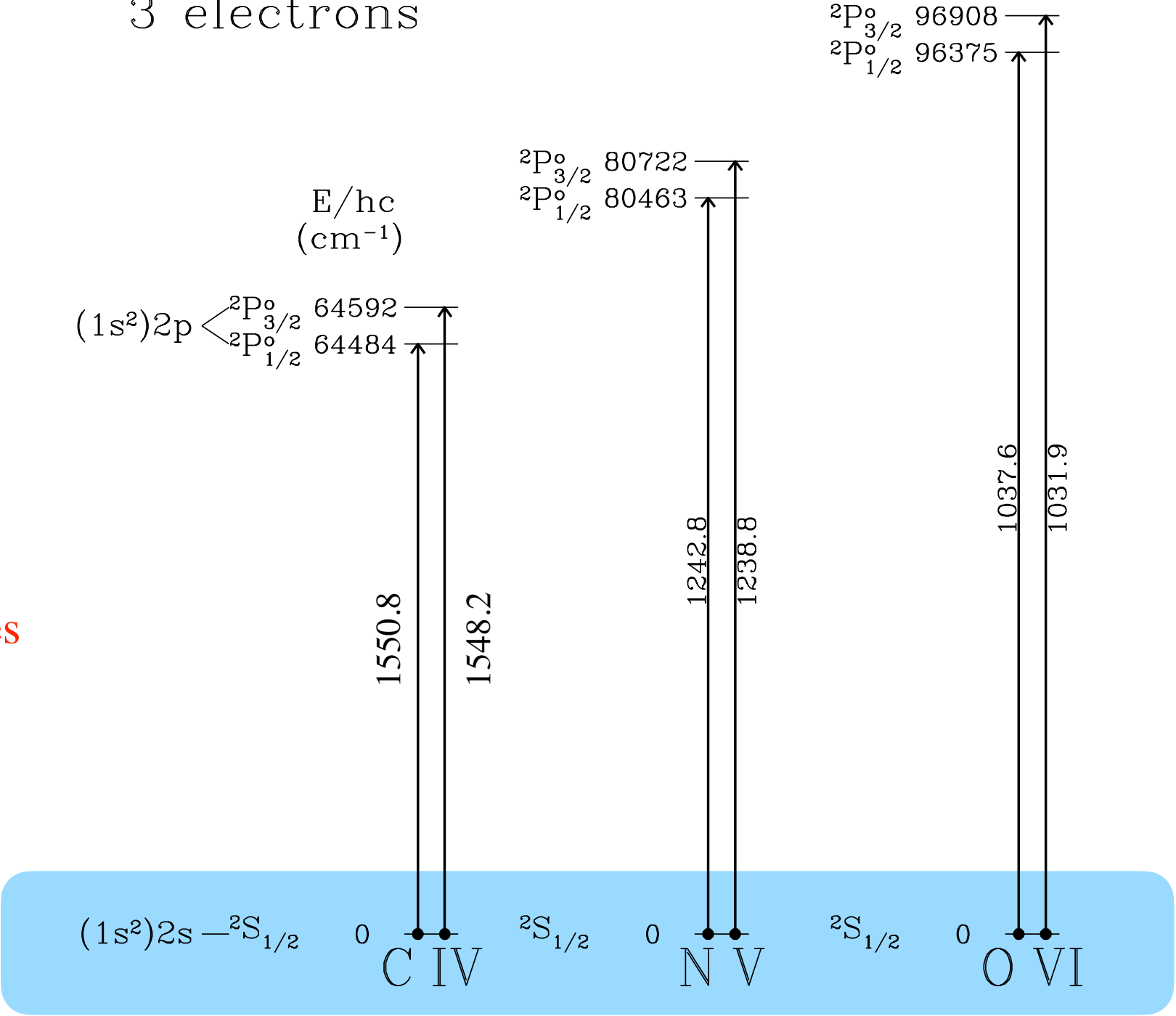
- 3 electrons (Lithium-like ions)

--- (13.6 eV)/hc = 109692 cm⁻¹ -----

3 electrons

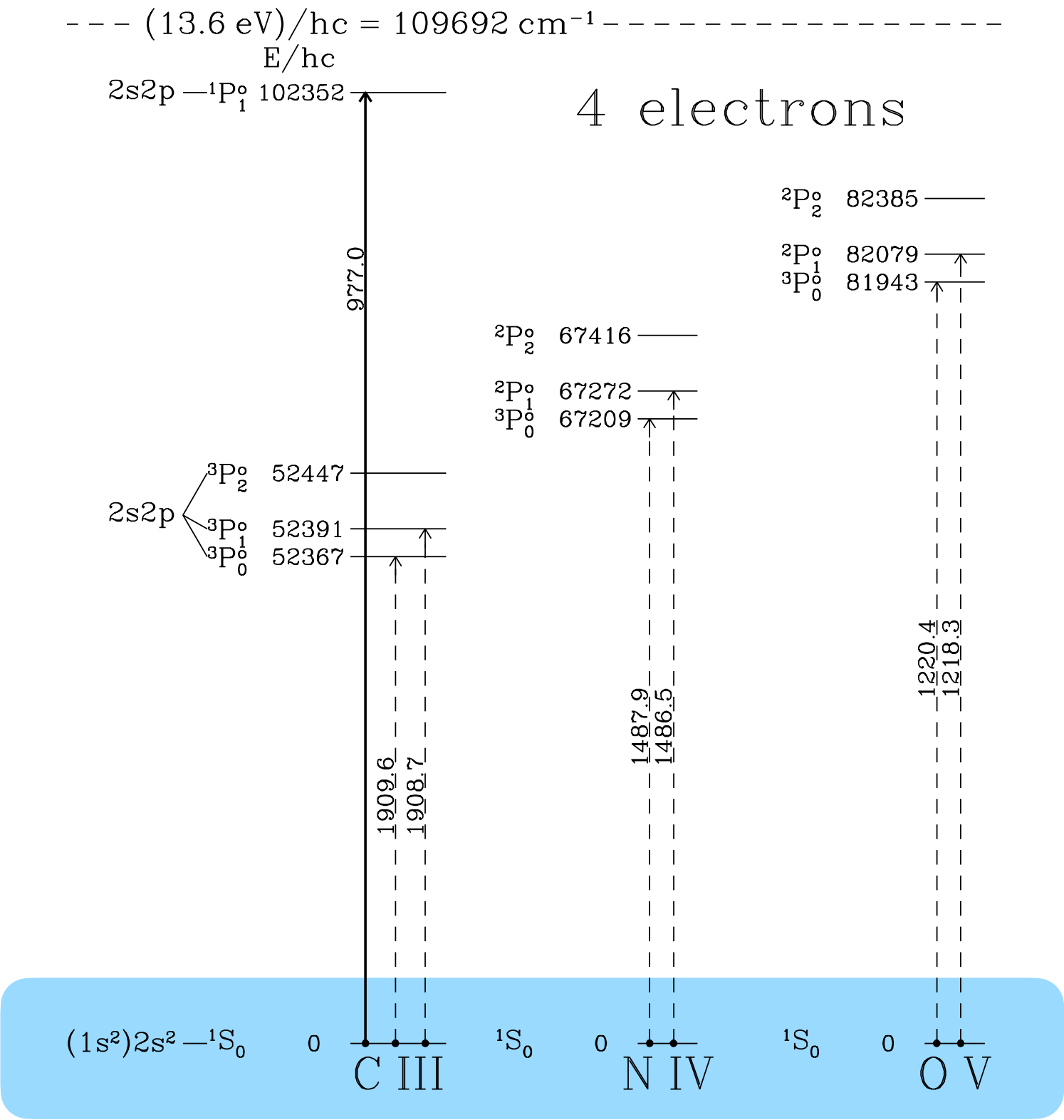
resonance lines

ground state



- 4 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden



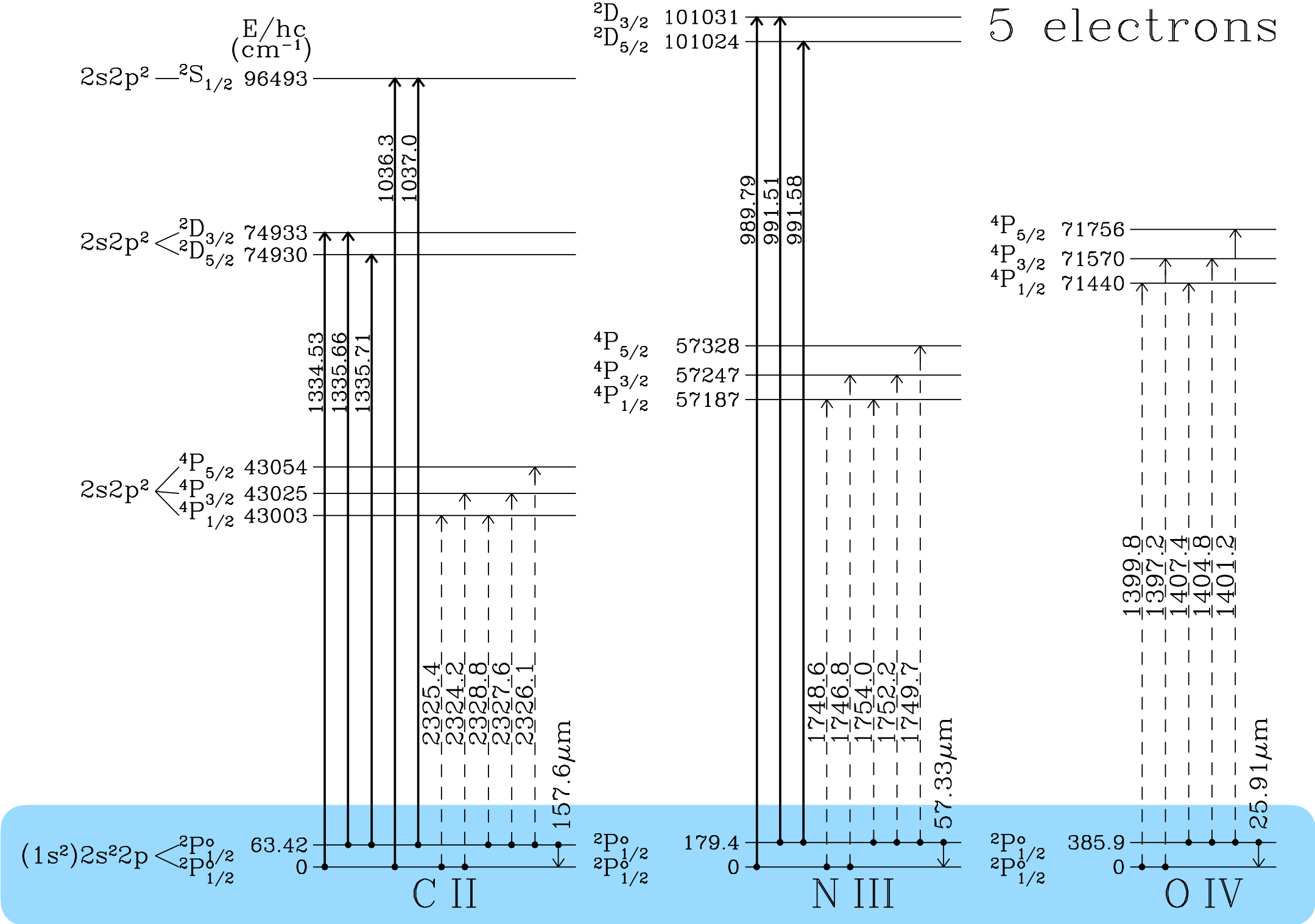
- 5 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

--- (13.6 eV)/hc = 109692 cm⁻¹ ---

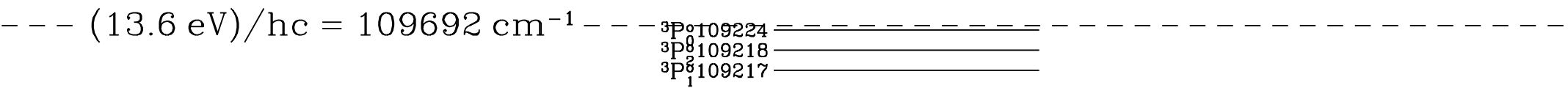
5 electrons

forbidden
lines

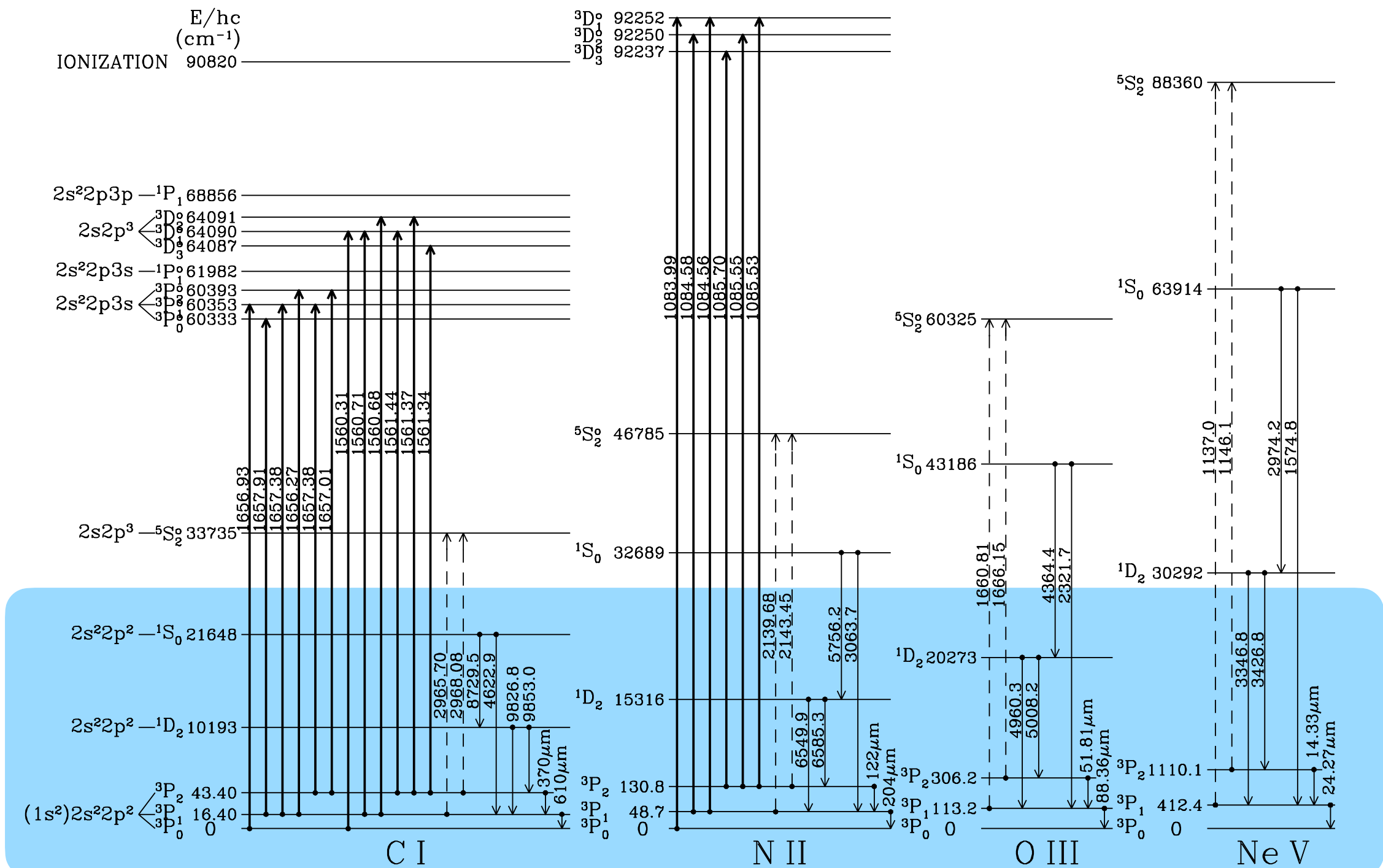


- 6 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden



6 electrons



forbidden
lines

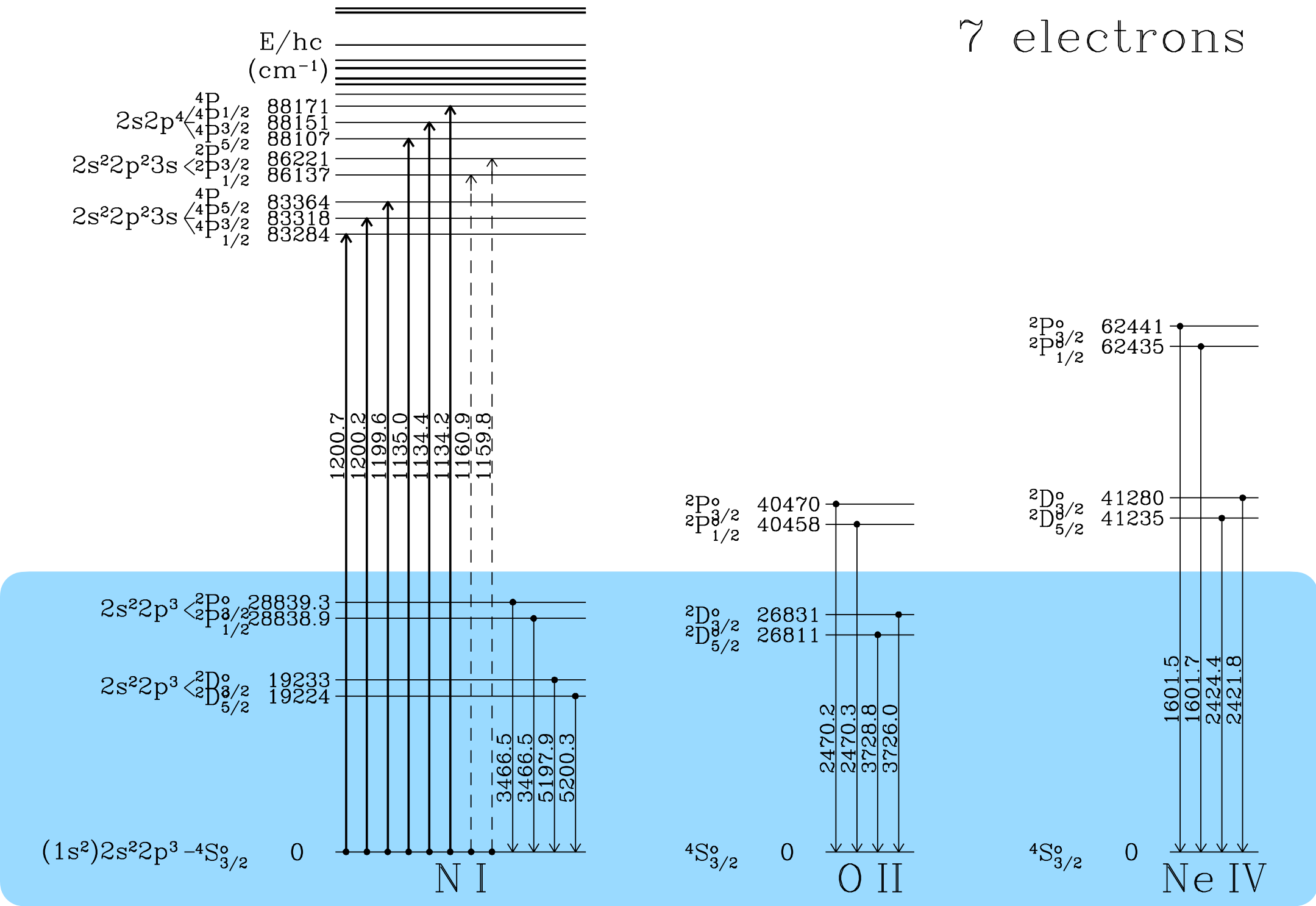
- 7 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

--- $(13.6 \text{ eV})/hc = 109692 \text{ cm}^{-1}$ ---

7 electrons

forbidden
lines



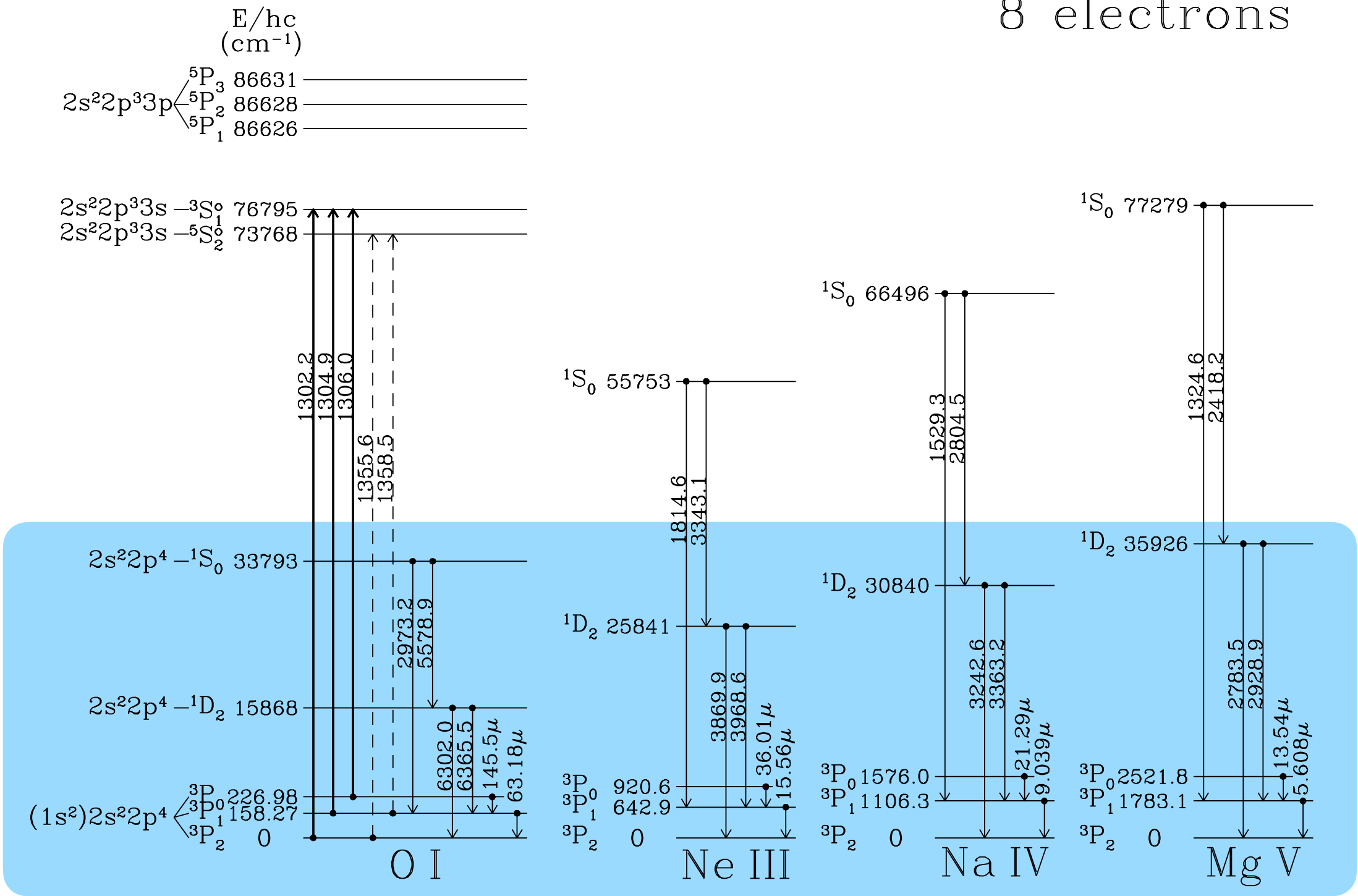
- 8 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

--- $(13.6 \text{ eV})/hc = 109692 \text{ cm}^{-1}$ ---

8 electrons

forbidden
lines

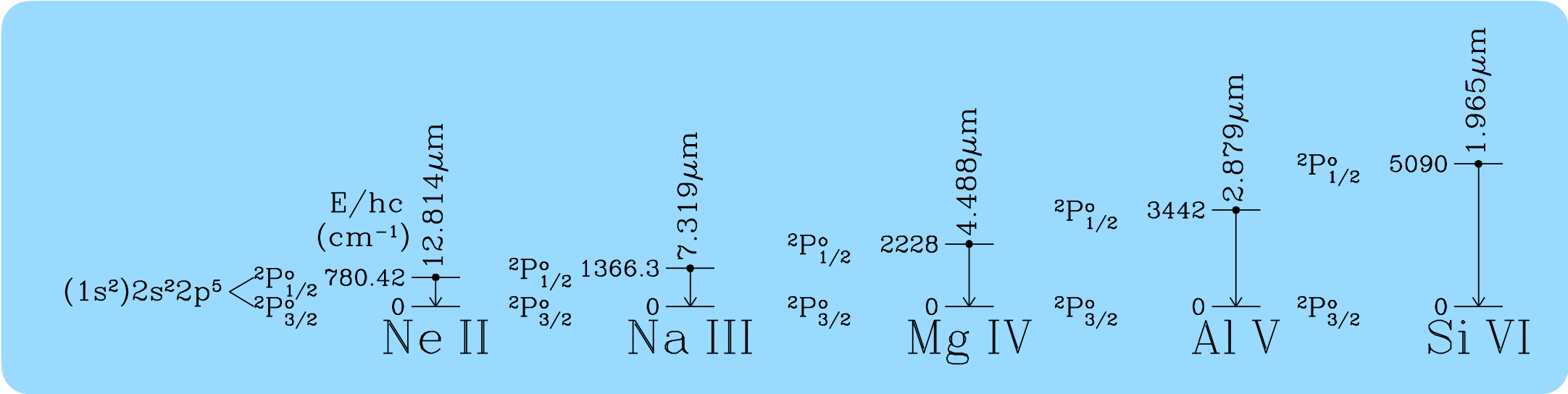


- 9 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

--- (13.6 eV)/hc = 109692 cm⁻¹ -----
9 electrons

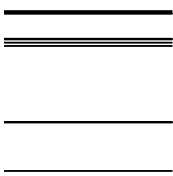
forbidden
lines



- 11 electrons

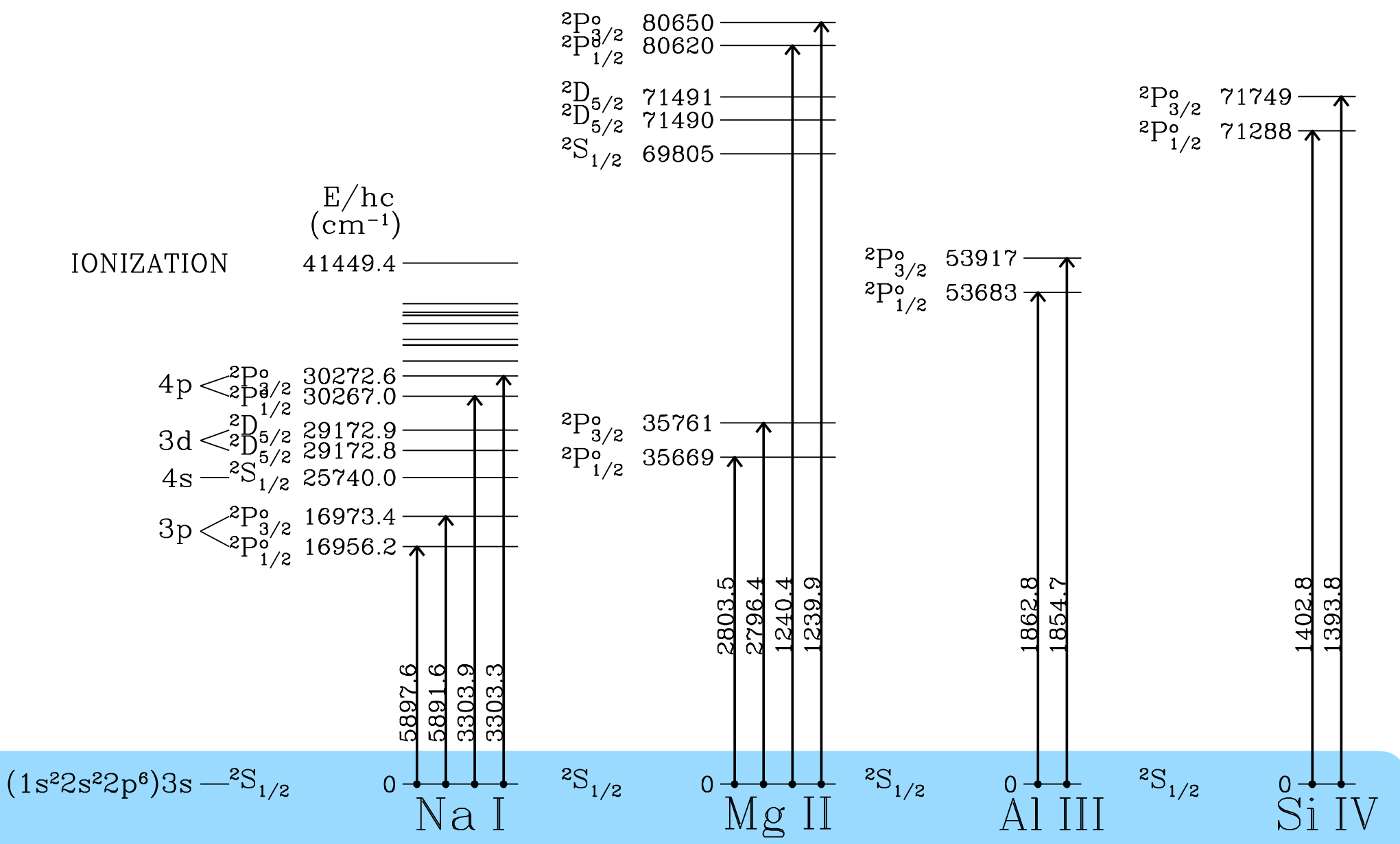
--- $(13.6 \text{ eV})/hc = 109692 \text{ cm}^{-1}$ ---

11 electrons



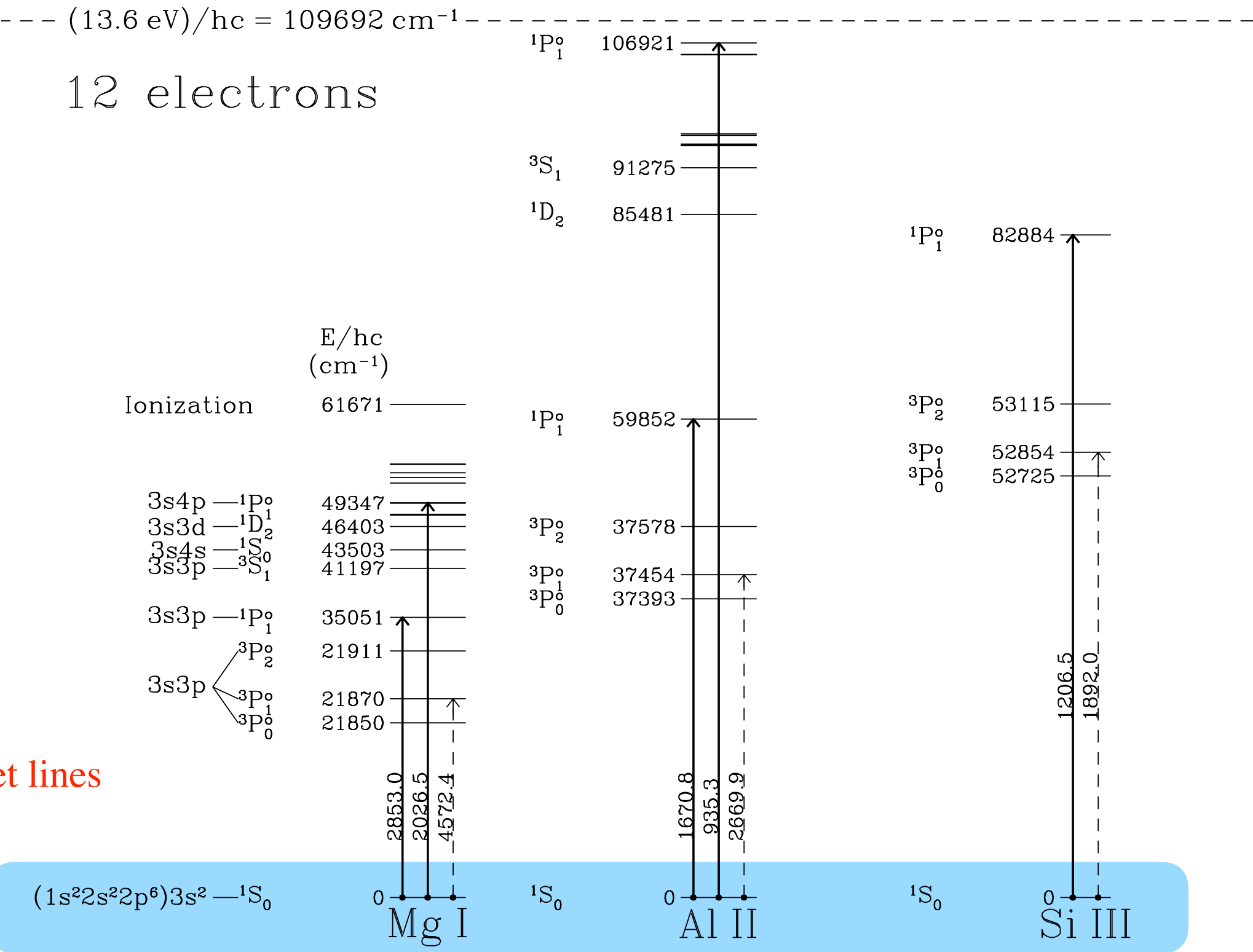
resonance lines

ground state



- 12 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden



resonant doublet lines

ground state

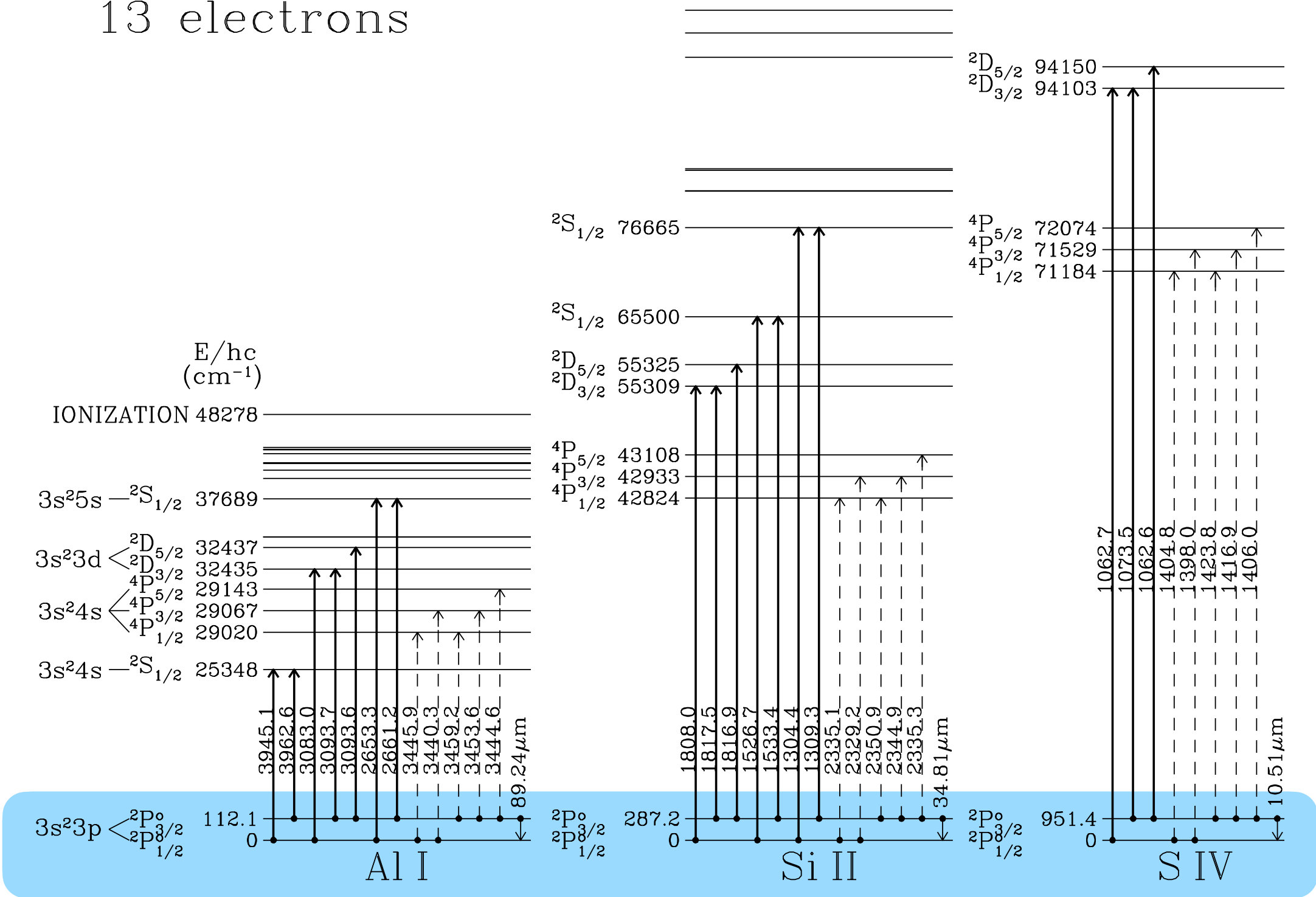
- 13 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

--- (13.6 eV)/hc = 109692 cm⁻¹ ---

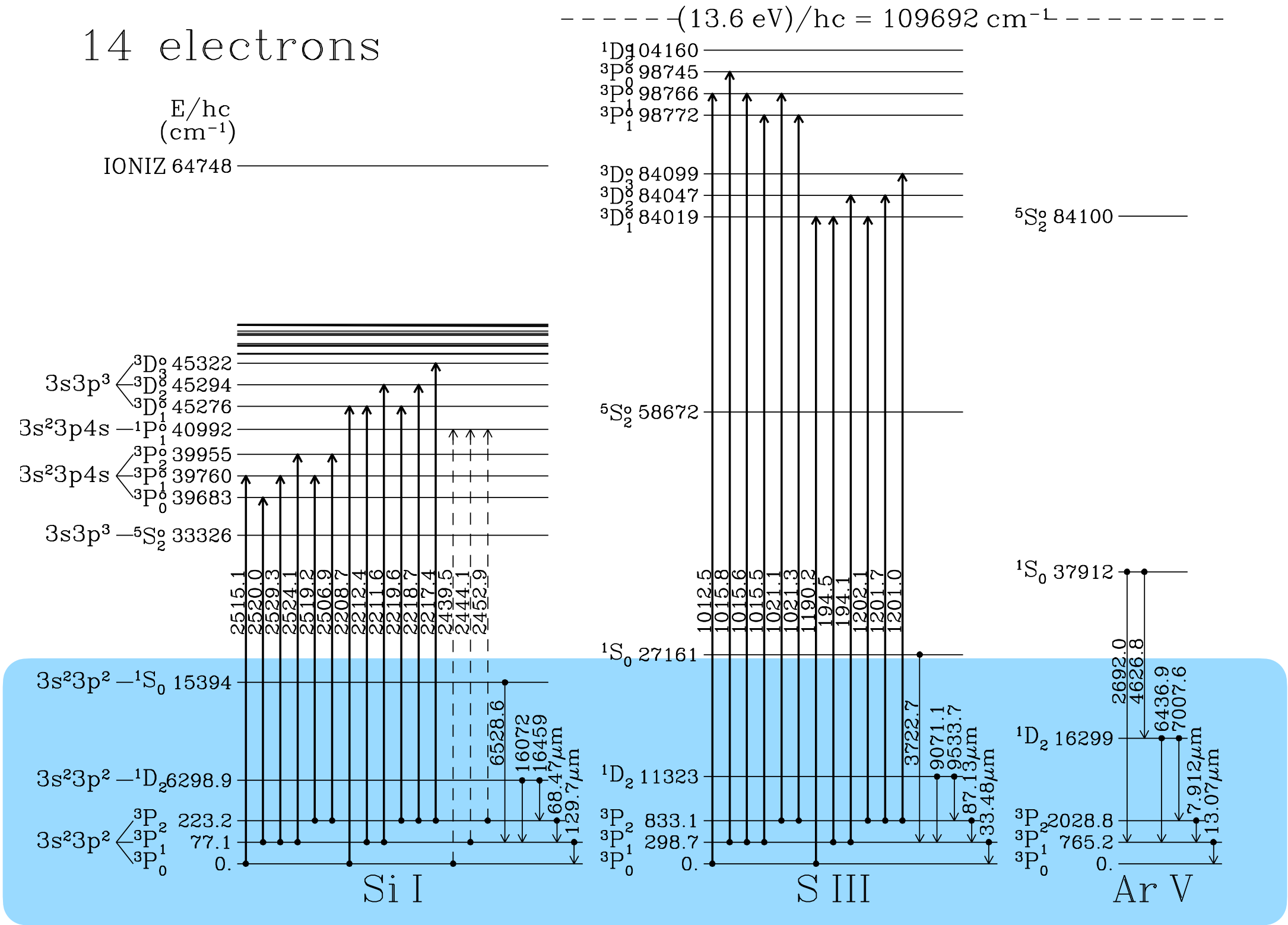
13 electrons

forbidden
lines



Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

14 electrons

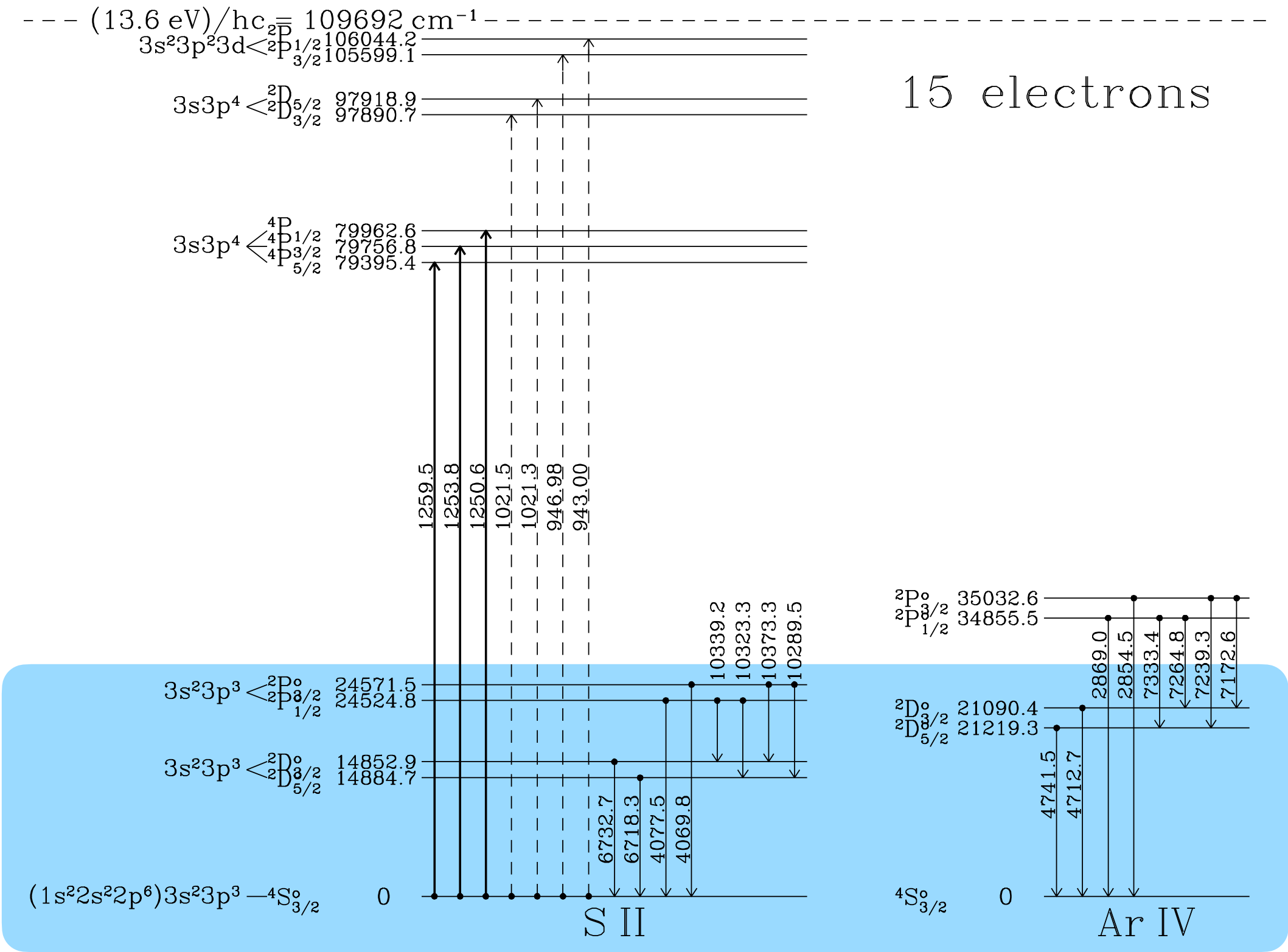


- 15 electrons

Upward heavy: resonance, Upward Dashed: intercombination
Downward solid: forbidden

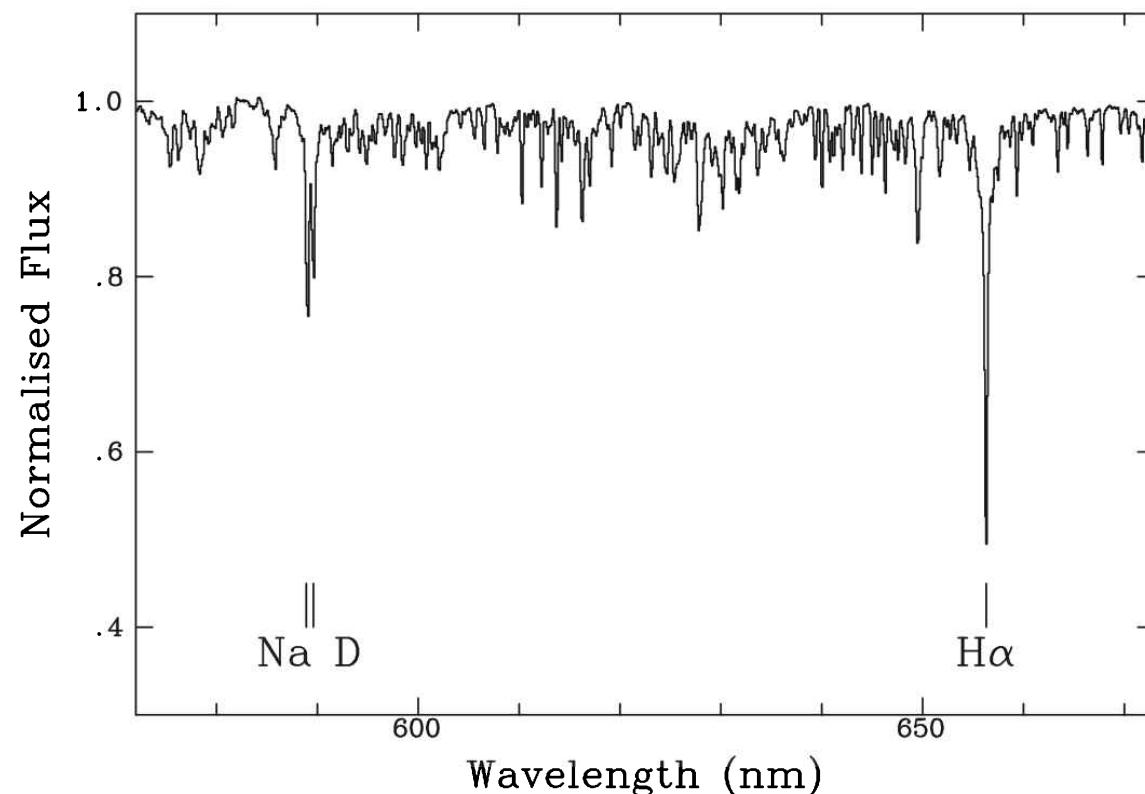
15 electrons

forbidden
lines

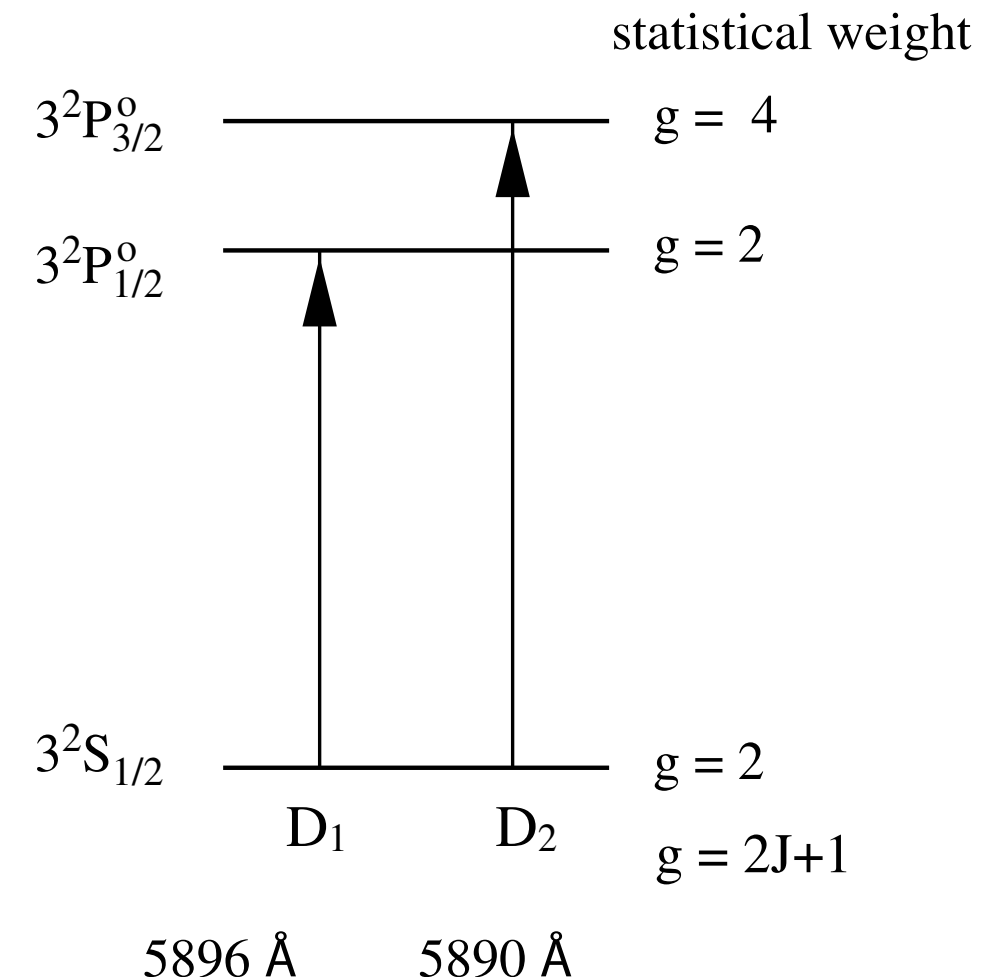


[Alkali Atoms]

- Alkali atoms: Lithium, sodium, potassium and rubidium **all have ground state electronic structures which consist of one electron in an s orbital outside a closed shell.**
- Sodium (Na) : Sodium has $Z = 11$ and a ground state configuration of $1s^2 2s^2 2p^6 3s^1$.



A solar spectrum reflected from the Moon just before a lunar eclipse taken at the University of London Observatory (S.J. Boyle).



Na D lines:

D₁ 5896 Å line : $3^2S_{1/2} - 3^2P_{1/2}$

D₂ 5890 Å line : $3^2S_{1/2} - 3^2P_{3/2}$

- Ca II (potassium-like calcium)

H 3968.47 Å line : $4\ ^2S_{1/2} - 4\ ^2P_{1/2}^o$

K 3933.66 Å line : $4\ ^2S_{1/2} - 4\ ^2P_{3/2}^o$

8498.0 Å line : $4\ ^2P_{3/2}^o - 3\ ^2D_{3/2}$

8542.1 Å line : $4\ ^2P_{3/2}^o - 3\ ^2D_{5/2}$

8662.1 Å line : $4\ ^2P_{1/2}^o - 3\ ^2D_{3/2}$

(Note that $^2P_{1/2}^o - ^2D_{5/2}$ is forbidden because $\Delta J = 2$.)

- Mg II (sodium-like magnesium)

2802.7 Å line : $3\ ^2S_{1/2} - 3\ ^2P_{1/2}^o$

2795.5 Å line : $3\ ^2S_{1/2} - 3\ ^2P_{3/2}^o$

1240.4 Å line : $3\ ^2S_{1/2} - 4\ ^2P_{1/2}^o$

1239.9 Å line : $3\ ^2S_{1/2} - 4\ ^2P_{3/2}^o$

- C IV (lithium-like carbon)

1550.8 Å line : $2\ ^2S_{1/2} - 2\ ^2P_{1/2}^o$

1548.2 Å line : $2\ ^2S_{1/2} - 2\ ^2P_{3/2}^o$

- N V (lithium-like nitrogen)

1242.8 Å line : $2\ ^2S_{1/2} - 2\ ^2P_{1/2}^o$

1238.8 Å line : $2\ ^2S_{1/2} - 2\ ^2P_{3/2}^o$

- O VI (lithium-like oxygen)

1037.6 Å line : $2\ ^2S_{1/2} - 2\ ^2P_{1/2}^o$

1031.9 Å line : $2\ ^2S_{1/2} - 2\ ^2P_{3/2}^o$

[Zeeman Effect]

- **Zeeman effect**

the splitting of atomic energy levels in the presence of an external magnetic field.

- **Zeeman effect at 1420MHz (21.1 cm)**

(1) The upper level of the 1420 MHz hyperfine transition of hydrogen is the $F = 1$ state. Its angular momentum has three possible projections onto a given axis. The three states are

$$F_z = m_F \hbar \quad (m_F = -1, 0, +1)$$

The three states have the same energy (i.e., they are degenerate).

However, if the atoms are immersed in an external magnetic field, the three states take on different energies. This is a consequence of the interaction of the magnetic moment of the hydrogen atom with the external field.

$$\begin{aligned} E_{\text{pot}} &= -\boldsymbol{\mu} \cdot \mathbf{B} \simeq g_e \frac{e}{2m_e} \mathbf{F} \cdot \mathbf{B} \\ &= g_e \frac{e\hbar}{2m_e} m_F B \end{aligned}$$

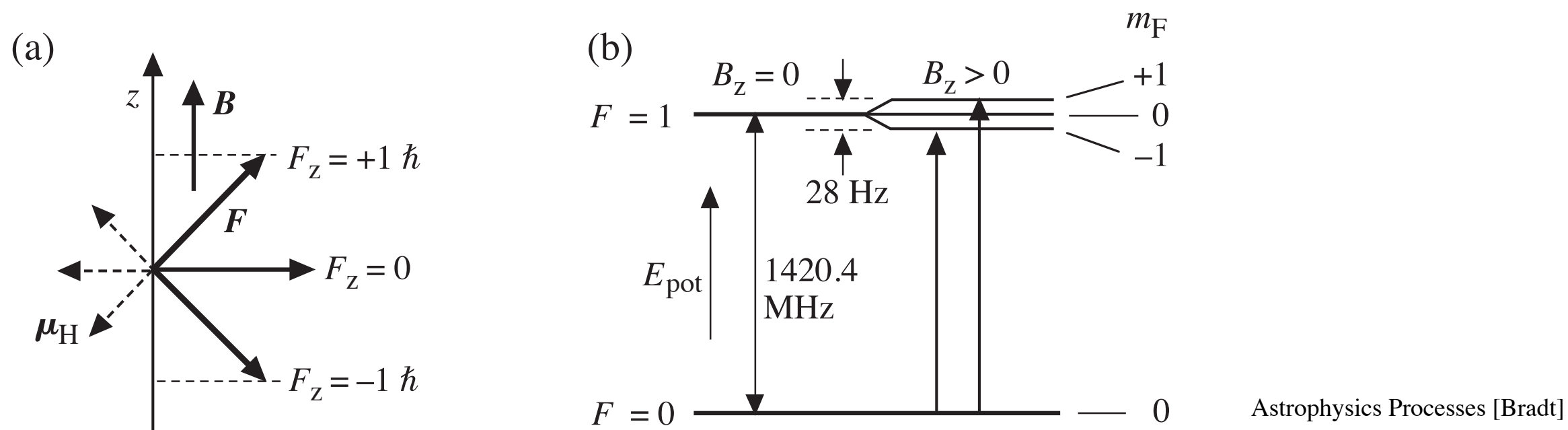
$\mu \simeq \mu_e$ because μ arises almost totally from the electron.

The separation between the +1 and -1 levels is

$$\Delta E_{\text{pot}} = E_{\text{pot}}(m_F = +1) - E_{\text{pot}}(m_F = -1) = g_e \frac{e\hbar}{m_e} B$$

(2) In the $F = 0$ ground state, the total angular momentum is zero. Therefore, the magnetic moment is zero and no Zeeman splitting occurs in the ground state.

Photons carry one unit of angular momentum $\mathbf{J}$. The allowed transitions are from the $F = 0$ ($m_F = 0$) to the $F = 1$ ($m_F = +1$) or $F = 1$ ($m_F = -1$).



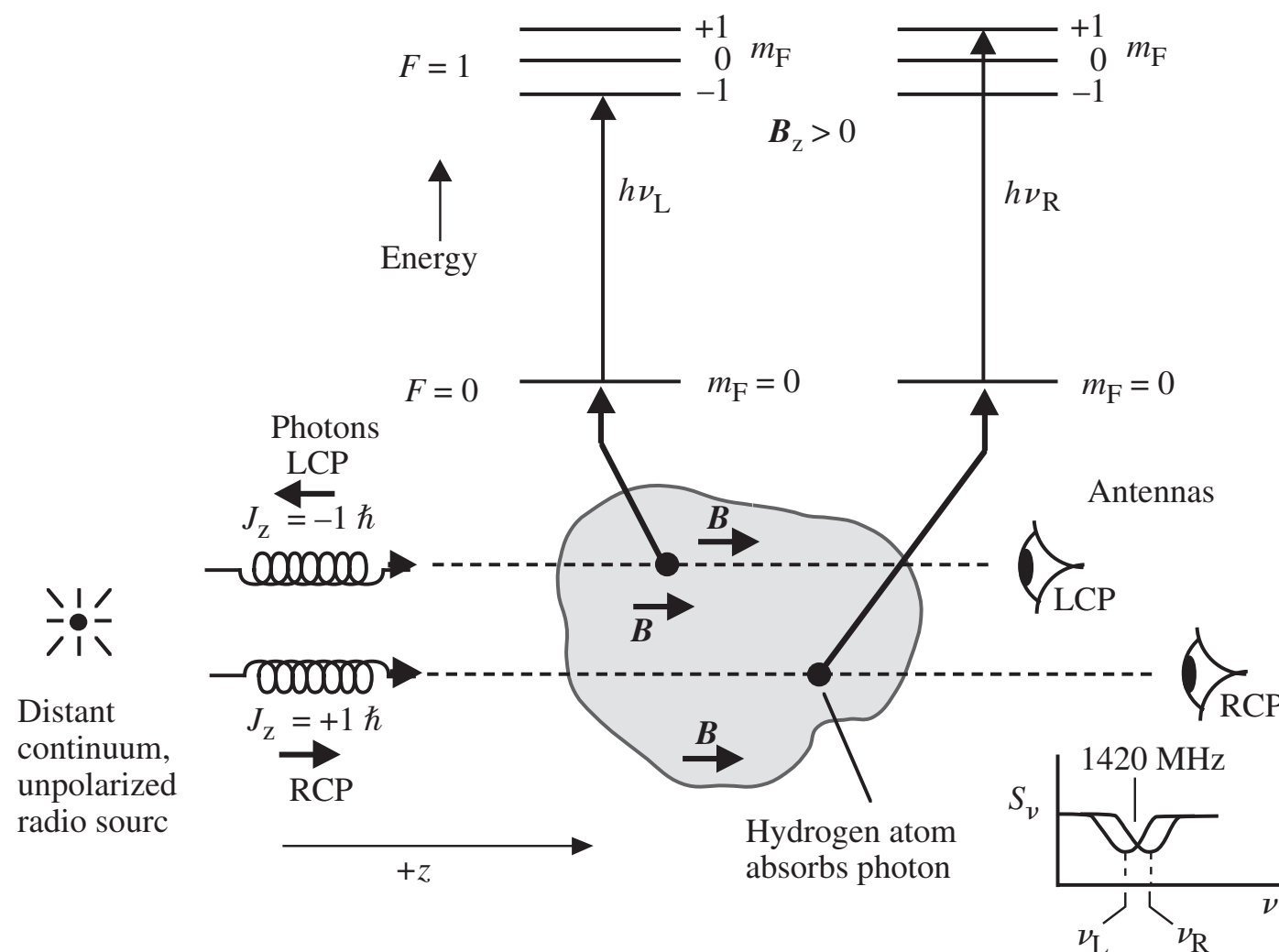
Energy levels for Zeeman absorption by hydrogen atom in an external magnetic field $\mathbf{B}$. The z -axis is taken to be in the direction of $\mathbf{B}$.

- (a) The three allowed projections ($m_F = +1, 0, -1$) of the angular momentum for the $F = 1$ state. The atomic magnetic moment μ_H , which is directed opposite to the angular momentum, interacts with the magnetic field to perturb the energy state.
- (b) Energy levels showing (left) the 1420-MHz transition for $B = 0$ and (right) the Zeeman splitting of the upper energy level into the three sublevels for $B_z > 0$. The two allowed transitions to the upper states are shown; they differ in frequency by 28 Hz for $B_z = 1 \text{ nT} (= 10\mu\text{G})$.

The typical magnetic field strength in the ISM is $\sim 3\text{--}6 \mu\text{G}$ near the Sun, and is higher in dense clouds and near the Galactic center ($\sim 10\text{--}100 \mu\text{G}$).

In quantum theory, a single photon is always circularly polarized with $S = \pm \hbar$ (no state with $S = 0$). Linearly polarized photon state is a superposition of a pair of circularly polarized photons.

A magnetic field of order $10 \mu\text{G}$ shifts the frequency by about one part in 10^8 of the hyperfine splitting. This shift is much smaller than the frequency shift $v/c \sim 10^{-5}$ due to a radial velocity of a few km s^{-1} , and it would be nearly impossible to detect, except that it leads to a shift in frequency between two circular polarization modes. The Zeeman effect in H I 21 cm can therefore be detected by taking the difference of the two circular polarization signals. This technique has been used to measure the magnetic field strength in a number of H I regions.



$$\nu_R - \nu_L = \frac{eB_z}{2\pi m_e} = 28.0 \left(\frac{B_z}{1.0 \text{ nT}} \right) \text{ Hz}$$

Left circularly polarized (LCP) photons are absorbed at a slightly lower frequency.
Right circularly polarized (RCP) photons are absorbed at a slightly higher frequency.
(Bradt: Astrophysics Processes)

See also *Heiles & Robishaw (2009)*